



Boundary spanning through external technology acquisition: The moderating role of star scientists and upstream alliances

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ABSTRACT

Building on the theories of *knowledge recombination*, we argue that the external acquisition of technologies acts as a *boundary spanning* mechanism that impacts on the recipient firm's subsequent technology development. The effect is moderated by two additional mechanisms, namely the retention of star scientists and the experience in upstream strategic alliances. We tested our hypotheses on a sample of 6208 USPTO patented technologies acquired by 350 biotechnology firms over the period 1980–2012. Findings reveal an inverted U-shaped effect of the pioneering nature of the acquired technology on firm's subsequent developments, in terms of (self) citations of firm's subsequent patents to the acquired one. Moreover, the main relationship is negatively moderated by the retention of star scientists, while the experience in upstream alliances demonstrates to be a positive moderator.

1. Introduction

In high-technology industries, firms' growth, development and, in the end, their survival depend on the ability to introduce new technologies and products over time (Kogut and Zander, 1992; Christensen, 2000; Nicholls-Nixon and Woo, 2003; Nerkar and Roberts, 2004). The so-called "technological responsiveness" (Nicholls-Nixon and Woo, 2003:651), hence, the ability to continually upgrade their technological know-how and capabilities (Nerkar and Roberts, 2004) have been the object of mounting literature based on the *knowledge recombination* construct – the process to re-combine previous pieces of knowledge in new forms and outputs (Nelson and Winter 1982; Henderson and Clark 1990; Kogut and Zander, 1992; Grant, 1996; Fleming 2001; Savino et al., 2017).

Knowledge recombination originates from firms' technology sourcing activity which is inherently multidimensional as it encompasses both the search and activation of internal pieces of knowledge and the search and use of external pieces of knowledges through different sourcing mechanisms (Hagedoorn and Schakenraad, 1994; Nicholls-Nixon and Woo, 2003; Nerkar and Miceli, 2016). However, knowledge recombination efforts are not costless due to the inherent 'stickiness' of information (Von Hippel, 1994; Szulanski 1996), the raise of internal boundaries for appropriation over in-house developed knowledge (Arrow 1962; Williamson, 1991), and because "the costs of coordinating and processing

information increase with distance" (Nerkar and Miceli, 2016:2; March and Simon, 1958; Cyert and March 1963).

Mounting research have therefore tried to explain the organizational search heuristics (Nelson and Winter 1982), demonstrating that only searching for knowledge outside the current domains (*boundary spanning*, Nerkar and Miceli, 2016) allows reaching a greater impact in terms of entering new markets, firm's technological repositioning and subsequent technological development (Levinthal and March 1993; Helfat, 1994; Stuart and Podolny, 1996; Martin and Mitchell, 1998; Rosenkopf and Nerkar, 2001; Fleming and Sorenson, 2004). Hence, despite of firm's tendency towards 'local search' processes (March and Simon, 1958; Cyert and March 1963), only balancing between exploitative (local) and explorative (distant) search behavior (Cohen and Levinthal 1990) allows firms to optimize their *knowledge recombination* efforts.

It is this interest in exploring the direction and impact of boundary spanning activities on the technological development of the focal firms that motivates the present study. By embracing the Nerkar and Miceli's (2016) interpretation of *boundary spanning* as an empirical method of dichotomizing *exploration* versus *exploitation* (Henderson and Cockburn 1994; Almeida and Kogut 1999; Ahuja and Lampert 2001; Rosenkopf and Nerkar 2001; Nerkar and Roberts 2004; Phene et al., 2006; Tallmann and Phene, 2007), we analyze the acquisition of technology characterized by a large extent of pioneering nature (*pioneering patent acquisition*) as a means for spanning both organizational and

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technological boundaries. Tapping to other firms' knowledge base, such as in the case of technology acquisition, allows the firms to access technology that is outside the current capabilities/domains (Nerkar and Miceli, 2016). In addition, in the specific case of pioneering patent acquisition, such experimentation, may provide the opportunity for the recipient firm to "circumvent the dangers of the propinquity trap" (Ahuja and Lampert, 2001:529; Brown, 1991) and therefore have a greater impact on subsequent technological development.

Empirically, we set up our research in a science-based industry, the biotechnology sector, where the high complexity and dynamism of the competitive scenario pose greater challenges to managers to continually upgrade their technological capabilities and, hence, to recombine knowledge across boundaries (Parmigiani, 2007; Nerkar and Miceli, 2016). Following patent-based studies (e.g. Ahuja and Lampert, 2001; Rosenkopf and Nerkar, 2001; Shane, 2001), we investigated how the pioneering patent acquisition affects subsequent technological developments (firm's subsequent patents' self-citations to the acquired patent). However, pioneering patent acquisition is not the only *boundary spanning* activity that firms can pursue, especially when we look at this Industry. These activities may act at different levels, and their interaction and interplay may impact the results of the overall organizational *boundary spanning* strategy. In details, alongside the acquisition of pioneering patent, we analyzed the concurrent effect of two boundary spanning activities, which are usually deployed in the biotechnology sector, at the individual and the alliance levels. We specifically refer to: 1) leveraging internal key employees (star scientists' retention); 2) pursuing a different type of knowledge recombination through inter-organizational mechanisms (upstream alliance experience); we consider both as moderators of our main relationship. Indeed, both employing star scientists - those individuals who have demonstrated superior levels of productivity and visibility within the scientific community (Zucker et al., 1998; Lacetera et al., 2004; Kehoe and Tzabbar, 2015) - and establishing upstream alliances - including grant, research, and R&D alliances (Powell et al., 1996; Hess and Rothaermel, 2011; Grigoriou and Rothaermel, 2017) - are common practices in the biotechnology sector (Zucker et al., 1998; Hess and Rothaermel, 2011; Tzabbar et al., 2013).

Hence, based on a database of 6208 patents whose ownership rights were transferred to 350 different biotechnology firms, our results reveal that the pioneering nature of an acquired patented technology exerts an inverted U-shaped effect on the extent to which a firm exploits this technology for future technological developments. This effect is negatively moderated by the employment of star scientists, while it is positively affected by the establishment of upstream alliances.

Based on our study, we therefore extend the extant literature in several ways: i) by contributing to the empirical use of the *boundary spanning* construct (Rosenkopf and Nerkar, 2001; Nerkar and Miceli, 2016), focusing on one of the mechanisms for spanning both organizational and technological boundaries (*pioneering patent acquisition*) and investigating its impact on firms' subsequent technological output (*Hp1*); ii) by contributing to the emerging micro-level theory of knowledge recombination (Fleming et al., 2007; Reinholt et al., 2011; Paruchuri and Eisenman, 2012; Aalbers et al., 2013; Gruber et al., 2013), which posits that individuals are at the locus of the knowledge recombination activity as "through their knowledge and skills technological components get recombined and potentially path-breaking inventions created" (Gruber et al., 2013:838) (*Hp2*); iii) by enriching the firm level approach of the *knowledge recombination* in technological innovation, focusing on firms' performance of pursuing multifaceted *knowledge recombination* efforts, including technology acquisitions and upstream alliances (Helfat 1994; Rosenkopf and Nerkar 2001; Miller et al., 2007) (*Hp3*). In sum, by investigating the relationship between the pioneering nature of the acquired technology and its exploitation by the acquiring firm, and how this relationship is moderated by two concurrent *boundary spanning* activities acting at different levels (i.e., individuals and alliances), we also enrich the innovation management

literature by analyzing and discussing innovation across multiple levels (Gupta et al., 2007). In this way, we offer an original perspective, accounting for interaction mechanisms, that allows to better understand the effects of *boundary spanning* activities on firm's overall innovation strategy.

The remainder of the paper is structured as follows. In the next section, we review the theoretical background and develop our hypotheses. We then present our research methodology and outline our results. We conclude the paper by discussing the results and main implications, as well as describing limitations of the paper that may spur further research.

2. Theory and hypotheses

2.1. Boundary spanning strategies and impact

Several authors have inquired how and under what circumstances firms can make the transition to a new technology, reposition itself technologically and therefore embrace a technological change (Stuart and Podolny, 1996; Nagarajan and Mitchell, 1998; Ahuja and Lampert, 2001; Nicholls-Nixon and Woo, 2003; Nerkar and Roberts, 2004). According to them, "technological responsiveness" (Nicholls-Nixon and Woo, 2003:651), which means the ability to upgrade skills and capabilities, is possible only if firms assimilate a range of technological inputs into novel combinations (Nerkar and Roberts, 2004). Hence, starting from Schumpeter (1934) the *knowledge recombination* - the process to re-combine previous pieces of knowledge in new forms and outputs - is considered lying at the very heart of firms' ability to create new product and new technologies, which are deemed essential for firm's development, growth, and survival (Nelson and Winter 1982; Henderson and Clark 1990; Kogut and Zander, 1992; Grant, 1996; Fleming 2001; Savino et al., 2017).

Knowledge recombination origins from firms' technology sourcing activity which is inherently multidimensional as it encompasses both the search and activation of internal as well as external pieces of knowledge (Hagedoorn and Schakenraad, 1994; Nicholls-Nixon and Woo, 2003; Nerkar and Miceli, 2016). This is particularly crucial as firms are increasingly in need of drawing on the knowledge stocks of other firms to face the limits and constraints of internal research and development. However, knowledge recombination efforts are not costless for several reasons, such as the inherent 'stickiness' of information (Von Hippel, 1994; Szulanski 1996), the presence of internal boundaries for appropriation over in-house developed knowledge (Arrow 1962; Williamson, 1991), and especially because "the costs of coordinating and processing information increase with distance" (Nerkar and Miceli, 2016:2; March and Simon, 1958; Cyert and March 1963).

Hence previous research, in the attempt to explain the organizational search heuristics (March and Simon, 1958; Nelson and Winter 1982), have proved the tendency that firms have to follow path-dependent trajectories, and search for solutions in the neighborhood of their existing expertise or knowledge (*local search*) (Helfat, 1994; Stuart and Podolny, 1996; Martin and Mitchell, 1998), given the fact they are more able to absorb external knowledge close to their existing knowledge base (Cohen and Levinthal 1990). Increasingly so, they have instead demonstrated that only moving away from these local attempts, and therefore searching for knowledge outside the current domains (*boundary spanning*, Nerkar and Miceli, 2016), it allows the company to reach a greater impact in terms of entering new markets, firm's technological repositioning and subsequent technological development (Levinthal and March 1993; Helfat, 1994; Stuart and Podolny, 1996; Martin and Mitchell, 1998; Rosenkopf and Nerkar, 2001; Fleming and Sorenson, 2004).

Balancing *boundary spanning* has become therefore crucial for firms to avoid and overcome the "core rigidities" (Leonard-Barton, 1995) or fall into "competency" (Levitt and March 1988) or "propinquity trap" (Ahuja and Lampert, 2001) and optimize the *knowledge recombination*

efforts (Nerkar and Miceli, 2016) along the two directions of *exploitation* and *exploration* proposed by March (1991).

Starting from Rosenkopf and Nerkar (2001), a bunch of studies have contributed to the interpretation and the analysis of *boundary spanning* as an empirical method of dichotomizing *exploration* versus *exploitation* (Henderson and Cockburn 1994; Almeida and Kogut 1999; Ahuja and Lampert 2001; Rosenkopf and Nerkar 2001; Nerkar and Roberts 2004; Phene et al., 2006; Tallmann and Phene, 2007) along several dimensions, including: organizational (Henderson and Cockburn, 1994; Rosenkopf and Nerkar, 2001), technological (Ahuja and Lampert, 2001; Nerkar and Roberts, 2004), and geographic (Almeida and Kogut, 1999; Phene et al., 2006; Tallmann and Phene, 2007). The dichotomy proposed have also guided the literature to focus on specific mechanisms for boundary spanning along the different dimensions and test their impact in terms of innovation/economic performance of the firm in the pursuit of its search processes. Authors have provided evidence on merger and acquisitions (Makri et al., 2010), licences (Laursen et al., 2010), alliances (Mowery et al., 1996; Rosenkopf and Almeida, 2003), hiring key external individuals (Almeida and Kogut, 1999; Rosenkopf and Almeida, 2003), but also losing key individuals (Agrawal et al., 2006; Corredoira and Rosenkopf, 2010) and demonstrated that in any circumstance new conduits for knowledge flow are opened and this increases a firm's interfirm learning potential leading to firm's technological development (Corredoira and Rosenkopf, 2010; Dokko and Rosenkopf, 2010).

2.2. Boundary spanning through patent acquisition

We consider patent acquisition as a boundary spanning mechanism allowing for exploring knowledge external to the firm (across the organizational dimension) and knowledge from distant technology domains (along the technological dimension), according to the Rosenkopf and Nerkar (2001)'s framework, as explained in the following paragraphs.

Along the organizational dimension, technology acquisitions, especially embedded in the form of patent acquisitions, have been systematically overlooked by extant research, in favor of all the remaining inter-organizational modes of boundary spanning. However, they are increasingly utilized and constitute a significant segment of markets for technologies (Serrano, 2010; Somaya, 2012; Caviggioli and Ughetto, 2013; De Marco et al., 2017; Figueroa and Serrano 2019), especially in science-based industries.

From the perspective of the recipient firm, patent acquisitions "not only convey the right over the use of a patented technology (as licensing does), but also the right to [...] use the technology, to enforce the patent against potential infringers, and the option to transfer all these rights, including the patent ownership, to another firm" (Figueroa and Serrano 2019:1602). Thus, there is a permanent change of ownership, which attributes to the receiving firm the full right to control all subsequent technology developments based on that technology (Tsai and Wang, 2008; Mudambi and Tallman, 2010), and this may encourage the receiving firms to invest in the new technological trajectory, which, in turn, may significantly shape its future competitive strategies (Blind et al., 2006). The potential impact of externally acquired patents on the subsequent technological evolution of the firm is further enhanced by their inherent characteristics. According to the available literature (Monk, 2009; Lichtenthaler, 2011; Caviggioli and Ughetto, 2013), the patents that are sold are most frequently marginal and non-core for the original patent holder, therefore, by definition, may they deviate from the technological trajectory the original firm focuses into and reveal potential value to firms operating in distant markets. Also, patent acquisition might arise when "the patent holder has abandoned a certain research trajectory or stopped operating within a certain industry or product domain" (Caviggioli and Ughetto, 2013:323; Monk, 2009). As a result, an opportunity for the "grafting" of external knowledge" (Ahuja and Katila, 2001: 197) into the internal base may ease the process of *knowledge recombination* leading to fruitful output of technological

development (e.g. new patents).

Along the technological dimension, our analysis investigates the specific case in which patent acquisition involves technologies characterized by a noticeable pioneering nature. In particular, Ahuja and Lampert (2001) considered pioneering technologies as those "that have no technological antecedents (i.e., they represent technologies that do not build on any existing technologies)" (p.523). By definition, these technologies diverge from both prior and current technological solutions (Dahlin and Behrens, 2005) and instead of trying to modify the current available processes, they focus on completely *de novo* solutions (Ahuja and Lampert, 2001). Hence, firms experimenting with highly pioneering technologies are more equipped to "circumvent the dangers of the propinquity trap" (p. 529) as they are pushed to adopt "a renewed look at cause-and-effect understanding, question existing cognitive structures, and uncover current competence deficiencies" (Zhou and Li, 2012:1093) and in the end, to embark on a distinct trajectory which should eventually yield to subsequent innovation development. However, the cognitive jump entangled by the acquisition and interpretation of those technologies, if, on the one hand, it may stimulate the technology development process (Van de Ven, 1986; Nonaka, 1994; Lin, 2003; Leone et al., 2016); on the other hand, it may require a huge effort in the deployment of novel capabilities with respect to the current existing competences retained by firms (Tushman and Anderson, 1986; Shane, 2001) and may call for greater absorption efforts for the recipient firm (Dahlin and Behrens, 2005; Phene et al., 2006). This circumstance is mostly true when facing patent acquisitions (while it is not in the case of licensing agreements, e.g. Leone et al., 2016), as they entail a permanent shift in the ownership right to the recipient firm, being the patent holder fully exempted from further commitment to the future technological development.

To illustrate this point, consider the case of the development of antisense drugs (i.e. compounds that inactivate genes that may cause specific disease). One of the pioneers, Gilead Science, sold its antisense patents portfolio to Ionis Pharmaceutical, after their decision to abandon that technology (Jarvis, 2006). The recipient firm was able to leverage the acquired patents to influence the subsequent technological evolution of the firm; as proof of this, subsequent patent filings of Ionis Pharmaceutical in the five years following the acquisition strongly rely on acquired patents, that were (self) cited 24.7 times on average. However, this was only possible because the company committed huge investments in terms of resources and co-specialized complementary assets for the exploitation of the acquired pioneering technology (Teece, 1986) to cope with the interpretation, use, development and deployment of these new pieces of knowledge (Tyre and Hauptman, 1992; Messeni Petruzzelli, 2015).

All that said, we argue that boundary spanning through the acquisition of patented pioneering technologies works up to a certain threshold because the excessive pioneering nature of these technologies poses greater challenges for the acquiring firms, as they need to adapt to the new technological paradigm and leverage it to innovate (Tushman and Anderson, 1986). Hence, we expect that:

H1. The pioneering nature of the acquired patented technology will be curvilinearly (inverted U-shape) related to the extent to which the acquiring firm exploits it for subsequent technological developments.

2.3. The moderating role of star scientists' retention and the experience in upstream alliances

Knowledge recombination efforts are multifaceted and very rarely they are pursued in isolation. Also, the knowledge recombination process happens at different levels which may interfere and may impact the results of the *boundary spanning* activity.

We therefore consider the effect of two additional mechanisms that contribute to the understanding of the overall process of *knowledge recombination* of interested firms. We refer to the retention of star

scientists and the experience in upstream strategic alliances (Hess and Rothaermel, 2011), as they represent common practices in science-based industries to unlock firms' internal innovation processes (Nickerson and Zenger, 2004; Jacobides and Billinger, 2006; Parmigiani, 2007; Parmigiani and Mitchell, 2009) through access of a variety of knowledge generation opportunities (Hess and Rothaermel, 2011).

We hence try to understand the moderating role of these common practices on the *boundary spanning* activated by the firm when pursuing pioneering patent acquisition. In other words, we argue these additional mechanisms act as moderator of our main relationship as they allow accumulating experience in codifying and transferring knowledge along the two dimensions, by affecting the processes of accessing, valuing and synthesizing external knowledge (technological-spanning capabilities) as well as accessing, connecting and partnering with external contributors (organizational-spanning capabilities) (Zollo and Singh, 2004).

2.3.1. The retention of star scientists

Star scientists are those scientists with superior levels of productivity in terms of publication activity and visibility within the scientific community (Zucker et al., 1998; Lacetera et al., 2004; Kehoe and Tzabbar, 2015). Their role is deemed fundamental on firms' innovativeness (Zucker et al., 1998; Fleming and Sorenson, 2004; Lacetera et al., 2004; Furukawa and Goto, 2006; Rothaermel and Hess, 2007; Azoulay et al., 2010; Almeida et al., 2011; Hess and Rothaermel, 2011; Oettl, 2012; Grigoriou and Rothaermel, 2014; Hohberger, 2016) for at least two reasons: on the one hand, they act as i) detector of new knowledge domains and promising technological opportunities (Fleming and Sorenson, 2004; Furukawa and Goto, 2006; Rothaermel and Hess, 2007); on the other hand they represent the ii) main contributor to the core knowledge base of the firm (Rothaermel and Hess, 2007), as they own the expertise and reputation needed to influence and drive the firm's technological and scientific development (Higgins et al., 2011).

According to the emerging micro-level theory of knowledge recombination (Fleming et al., 2007; Reinholt et al., 2011; Paruchuri and Eisenman, 2012; Aalbers et al., 2013; Gruber et al., 2013), star scientists can be also considered the favorable locus of the *knowledge recombination* activity of the firm given the fact that through their high level of scientific knowledge and productive skills, technological components get recombined and new solutions potentially be invented (Fleming and Sorenson, 2004; Gruber et al., 2013). Hence, in this vein, they can be considered as amplifier of "the *knowledge recombination* across technological boundaries, because they have acquired "a more abstract understanding of the technological problem-solving process and [...] [hence they] are more likely to have developed an ability in metal-earning that facilitates the assimilation of technological knowledge from other domains" (Gruber et al., 2013:839; Gibbons and Johnston 1974; Woolnough 1991, 1994). Also, according to Hohberger (2016), having a star directly involved in the generation of an invention, may spur firm's innovative performance, as they trigger the overall *knowledge recombination* of non-star scientists towards fruitful trajectories.

From this perspective, the ability of the firm to retain as many as possible star scientists may positively impact the prospects for firm's technology development on the basis of the pioneering nature of the externally acquired technology. The reasons for this is that firms, relying on star scientists, can better leverage their 'second-order competence' - the ability of a firm to create new knowledge through recombination of knowledge across (technological) boundaries (Rosenkop and Nerkar, 2001) - as they exploit their higher capabilities in understanding and further developing scientific knowledge to allow the firms more effectively accessing, valuing and synthesizing the external technologies characterized by a remarkable pioneering nature (Fleming and Sorenson, 2004). Based on this reasoning, we formulate the following hypothesis:

H2. The retention of star scientists positively moderates the relationship between the pioneering nature of the acquired patented technology and the extent to which the firm exploits it for future technological developments; in other words, the threshold level after which the pioneering nature of the acquired patented technology implies negative returns will be higher when firms employ more star scientists.

2.3.2. Experience in upstream alliances

Upstream strategic alliances (Powell et al., 1996; Hess and Rothaermel, 2011) are set up to drive explorative research activities in collaboration with external R&D partners who can contribute to enriching the knowledge base of firms (Koza and Lewin, 1998; Rothaermel and Deeds, 2004; Lavie and Rosenkopf, 2006; Lin et al., 2012). They are particularly useful and effective especially at the early stage of the research and development process (Rothaermel and Deeds, 2004) and when facing the challenge to renew technological leadership in high-paced competitive environments, because they allow the firms to be exposed to external knowledge opportunities (Koza and Lewin, 1998) for the creation of valuable and novel basic knowledge that otherwise would not be accessible to the innovating firms (Powell et al., 1996; Rothaermel and Deeds, 2004).

From the perspective of knowledge recombination, experience in upstream alliances can moderate the boundary spanning through pioneering patent acquisition in two ways. On the one hand, as firms tend to enter upstream alliances to generate new basic knowledge (see Koza and Lewin, 1998; Rothaermel and Deeds, 2004; Hess and Rothaermel, 2011), this engagement helps to broaden a firm's capacity to fully exploit externally acquired technology (Caloghirou et al., 2004). Also, as this type of agreement is usually signed to share and acquire tacit knowledge (Mowery et al., 1996; Hess and Rothaermel, 2011), compared to conventional contracts (e.g. licensing), it allows for the exchange of firm-specific technological capabilities that would otherwise be difficult to transfer between organizations. Hence, this circumstance further enhances the development of 'second-order competence' (Rosenkop and Nerkar, 2001), along the technological dimension, which is fundamental to fully understand, assimilate, and recombine the technologies endowed with greater pioneering nature that are acquired through patent contract. In addition, as the upstream alliances require building strategic relationship with external partners, involving the development of routines for negotiating with external partners, for codifying knowledge and transferring tacit knowledge (Zollo and Singh, 1997), which lays the foundation for building trust among partners (Gulati, 1998), experimentation with alliances, involving repeated interactions can improve this capability (Dyer and Singh, 1998) and therefore augment the second-order competence also along the organizational dimension. All taken together this may positively affect the impact of the pioneering technology acquisition on subsequent technological development of the recipient firm.

Hence, we formulate our third moderation hypothesis, as follows:

H3. Experience in upstream strategic alliances positively moderates the relationship between the pioneering nature of the acquired patented technology and the extent to which the firm exploits it for future technological developments; in other words, the threshold level after which the pioneering nature of the acquired patented technology implies negative returns will be higher as firms establish more upstream strategic alliances.

3. Methods

3.1. Setting and data

In order to test our hypotheses, we considered the biotechnology

industry¹ as a suitable setting for the purposes of this study, for several reasons. Firstly, this industry is characterized by a strong appropriability regime that favors patenting activity (e.g. Levin et al., 1987), and, in turn, the patent transfer practices that are in place among firms (Gans and Stern, 2010). As a matter of fact, patents in biotechnology play a crucial role in protecting the results of internal R&D activities (Zucker et al., 2002; Phene et al., 2006) and in promoting the trade of patent rights (Arora et al., 2001; Zucker et al., 2002). Secondly, biotechnology patents generally have a narrower scope and require more time and expertise to be developed, compared to patents in other sectors (Check Hayden, 2011). Thirdly, they must also comply with stringent standards and regulations, and consequently the adoption of opportunistic behaviors such as patent trolling and patent blocking (Chien, 2009) is less severe than in other industries. All these aspects considered, we are confident that, in our sample, firms tend to acquire patents to nurture their innovation processes rather than for strategic reasons. Finally, as regards the additional search mechanisms we look at, both star scientists (e.g. Zucker et al., 1998) and upstream strategic alliances (e.g. Hess and Rothaermel, 2011) constitute noteworthy innovation strategies for biotechnology firms, especially when they aim to develop new knowledge and competences.

To build our sample we drew on the full list of biotechnology firms included in the 2012 BioScan database² and we matched them³ to the corresponding patent assignment transactions⁴ extracted from the United States Patent and Trademark Office (USPTO) patents assignment database⁵ in the period 1980–2012. We focused on the USPTO not only because the United States is the biggest market for the patenting and trading of biotechnologies (e.g. Arora et al., 2001), but also because the U.S. market offers better conditions, thereby ensuring a higher level of reliability of the data. First of all, this makes it possible to downsize institutional differences that characterize and regulate the transaction of patented solutions across countries (Ginarte and Park, 1997); second, it provides the means to limit possible biases caused by patent trolling and patent blocking, since the U.S. patent system shows lower percentages of unused patents and higher percentages of licensed and sold patents with respect to the European and Japanese patent markets (Torrisi et al., 2016).

Following these data collection steps, we created a database with 6208 patents whose ownership rights were transferred to 350 different biotechnology firms. Our database shows that 4443 patents were acquired by large firms, corresponding about to the 72% of the sample. However, while we noticed that large firms tend to acquire more pioneering patents with respect to SMEs, no differences are present in the use of acquired patents (see Appendix B for a detailed analysis). The

average number of patents acquired by each firm is 17.74. Moreover, considering the 350 firms in the database, about 45% of them exclusively operate in the biotechnology sector, while the others are diversified into other sectors as the pharmaceutical one. Furthermore, we noticed that 53% of the firms are publicly listed. Finally, the median firm foundation year is 1993, highlighting the recency of firms in our database.

Afterwards, for each patent, bibliographic information was gathered from the USPTO database, including backward and forward citations, technological classes, and assignor type. Additionally, drawing from Scopus and the BioScan database, we integrated data on patent-transfer agreements with observations on star scientists and upstream strategic alliances, respectively. Finally, we obtained firm-level data by employing multiple sources, including BioScan, Securities and Exchange Commission (SEC) filings for publicly traded firms, Orbis (<https://orbis.bvinfo.com>), Compustat (<https://www.compustat.com>) and corporate websites. In the end, we were able to relate data on patent-transfer agreements, transferred patents, original and recipient firms, star scientists employed, and upstream strategic alliances undertaken by the recipient firms.

3.2. Variables

Dependent variable. The extent to which firms exploit acquired technologies to foster future technological developments and therefore develop proprietary innovations is our dependent variable and it was measured by the number of forward citations received by the acquired patents in the acquiring firms' other patented inventions (*SelfCit*) in the five years following the agreement year (Messeni Petruzzelli et al., 2015b). By doing so, we accounted for the fact that subsequent technological development, that is future proprietary technologies, builds upon the acquired patented technologies (Cattani, 2005). Actually, the self-citation represents the realization of successful boundary spanning. In other words, when a firm cites an acquired patent, it can be deemed as having successfully performed a boundary spanning activity, resulting in subsequent technological development leveraging and relying on the acquired technology. Indeed, the number of a patent's forward citations is a widely accepted indicator of its impact on subsequent technological progress (e.g. Trajtenberg, 1990; Rosenkopf and Nerkar, 2001) by revealing the successful transfer and employment of previous technologies (e.g. Jaffe et al., 1993). In addition, consistent with the empirical studies focusing on forward patent citations, we adopted a five-year citations lag, which is considered an appropriate interval to evaluate the impact of technologies (e.g. Nooteboom et al., 2007). Patent citations may be added both by the assignees and examiners (Alcácer and Gittelman, 2006), who also carefully check for the suitability of the previous ones, thus ensuring the inclusion of references only to prior useful technologies. Furthermore, it is worth mentioning that, in our case, both the citing and the cited patent belong to the same firm, thereby increasing our confidence in the consciousness of the firm when including the given citation.

Independent variable. The pioneering nature of the acquired patents (*PioneeringNature*) was evaluated on the basis of their reliance on prior technical art, as related to the number of backward patent citations referenced by the focal patent (Nerkar and Shane, 2007). In particular, we chose to rely on a continuous measure in order to account for the extent of the pioneering nature of acquired technologies, differently with respect to other studies that, instead, used only a binary variable to define if a technology is pioneering or not (e.g. Ahuja and Lampert, 2001; Van de Vrande et al., 2011). Prior citations have been widely adopted by previous studies to capture the technological base upon which patents are built (e.g. Jaffe et al., 1993). To favor the interpretation of the statistical analysis, we calculated *PioneeringNature* as the opposite of the number of focal patent's backward patent citations (i.e. zero minus the number of backward citations). Indeed, this means that a higher number of backward citations indicates a lower value of the

¹ The birth of the biotechnology industry is commonly set in 1974, when Stanley N. Cohen and Herbert W. Boyer filed an application to patent the recombinant DNA technique. Afterwards, in 1976, Boyer co-founded with Robert A. Swanson the Genentech company, which produced the first synthetic human insulin; this was the first genetically engineered human therapeutic drug approved by the U.S. Food and Drug Administration. Since then, the biotechnology industry has grown noticeably, to the point that more than 23 thousand biotechnology firms were reported as active in 2013, with an overall R&D expenditure exceeding 42 billion U.S. dollars (OECD, 2015).

² BioScan is one of the most comprehensive databases collecting information on strategic alliances in the biotechnology industry (e.g. Powell et al., 1996; Rothaermel and Deeds, 2004; Hess and Rothaermel, 2011).

³ To ensure data reliability, we performed a preliminary check, before the match, to validate the consistency of the names of the firms acquiring patents, since their identifiers may not be unique in the USPTO database (e.g. they can appear as full names, with alternative spellings, or even as acronyms).

⁴ Namely, the transfer of patent ownership interest (see Section 301 of the USPTO Manual of Patent Examining Procedure, available at <https://www.uspto.gov/web/offices/pac/mpep/index.html>, latest access on October 2019).

⁵ Available at: <https://www.uspto.gov/ebc/assignments.htm>, latest access on October 2019.

independent variable, which characterizes less pioneering technology. On the contrary, a more pioneering technology is characterized by a low level of backward citations, which corresponds to a higher value of the independent variable. Finally, it is not considered affected by retrospective and success bias either and offers a certain consistency for within-industry analyses (Dahlin and Behrens, 2005).

3.2.1. Moderating variables

Star Scientists retention. Two different moderating variables are introduced in our analysis, namely the retention of internal star scientists and the experience in upstream strategic alliances on the part of the recipient firm. To account for the retention of star scientists (*StarScientists*) we sampled from individual scholarly publication activity, consistent with previous studies suggesting that scientists authoring an outstanding number of academic publications are considered stars (e.g. Zucker and Darby, 1997; Zucker et al., 2002; Furukawa and Goto, 2006; Hess and Rothaermel, 2011). Furthermore, publishing activity has often been used to capture the scientific expertise of corporate scientists, providing evidence, among other things, of their contribution to the academic realm (Jong and Slavova, 2014). More specifically, we first extracted a list from the Scopus database of more than 200,000 scientists that published at least one article in academic journals while they were affiliated to one of the biotechnology firms listed in our sample (Hess and Rothaermel, 2011). After that, we identified the star scientists based on this criterion: an individual was deemed a star scientist, in a specific year t , if her/his average number of annual publications in a three-year timeframe (from $t-2$ to t) was equal to or greater than five in that specific year (see also Lacetera et al., 2004). Based on this operationalization, scientists were considered stars only in years characterized by recent remarkable publication activity; they did not qualify earlier, when they still had to reach star status, nor later, when their productivity, as well as their contribution, might have dropped. As such, the *StarScientists* variable is measured by the logarithm of the number of individuals considered as star scientists (plus one) affiliated with the acquiring firm in the year t . Since we are interested in the shift of the threshold of the inverted U-shaped relationship between *Pioneering* and *SelfCit* after which negative returns set in (i.e., the shift of the turning point of the relationship) and we did not expect this to change shape at any level of *StarScientists* (i.e. to become linear or U-shaped), we considered only the interaction between *StarScientists* and the linear term of *PioneeringNature* when we tested Hypothesis 2, and thus not including the interaction between *StarScientists* and the quadratic term of *PioneeringNature* (Aiken and West, 1991; Godart et al., 2014; Haans et al., 2016).

To offer a clearer picture of the context, in Fig. 1 we report the number of star scientists identified for each year. The graph shows a growing trend, highlighting the emerging relevance of star scientists in organizations' R&D activities.

Experience in Upstream Alliances. As concerns the other moderating variable, the experience in upstream strategic alliances (*Upstream*) was measured by the number of upstream alliances entered into by the

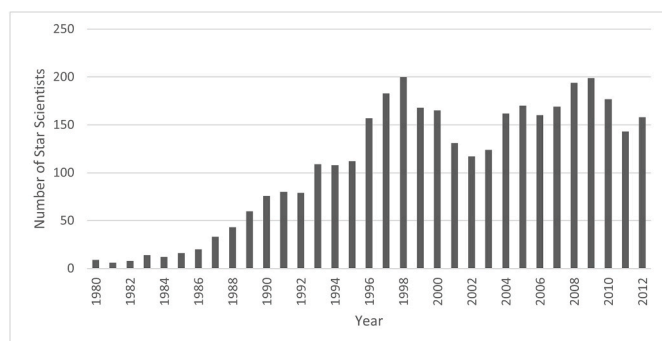


Fig. 1. Number of star scientists identified per year.

acquiring firms in the five-year span up to the patent acquisition agreement (Lavie and Rosenkopf, 2006; Rothaermel and Deeds, 2006). Specifically, following Koza and Lewin (2000), we collected the alliance announcements included in BioScan and we identified upstream relationships as those involving a knowledge generating R&D agreement (e.g. research and co-development alliances). In this way we obtained a list of 1821 upstream alliances pertaining to the biotechnology firms included in our sample. In Fig. 2, we show the number of upstream alliances per years identified. Actually, we noticed that number of upstream alliances rises since the beginning of 1990s, with a peak in the period 2004–2006. As for the previous moderator, we tested Hypothesis 3 interacting *Upstream* only with the linear term of *PioneeringNature*.

Control variables. The main effects of the moderating variables were also used as control variables. Additionally, other controls accounting for both the patents' and firms' characteristics were also included in the model, as listed below.

Patent-level variables. For each patent acquired, we controlled for the number of claims (*Claims*) (Lanjouw and Schankerman, 2004) and for the patent scope (*Scope*), measured as the number of different three-digit U.S. patent classes (Lerner, 1994; Nerkar and Shane, 2007), because previous literature showed they may both influence the impact of the patent. In addition, to account for peculiar trends that may affect older patents compared to more recent ones due to specific patent strategies (Fischer and Leidinger, 2014), we controlled for the age of the patent (*PatentAge*), measured as the number of years elapsing between the patent grant year and 2017. Furthermore, following Narin et al. (1997), we measured the patents' scientific references (*ScientificCit*) as the number of non-patent references, thus controlling for the linkages between science and technology. Finally, we also added dummy variables to account for the assignment of the patent to each of the three-digit patent classes pertaining to the biotechnology field (*BioClass*), as defined by USPTO⁶ (see Rothaermel and Thursby, 2007), or to other patent classes (*OtherClasses*) otherwise.

Firm-level variables. To control for the acquiring firm's characteristics, we included the effects of its size (*FirmSize*), calculated as the logarithm of the number of employees in the patent acquisition year, and its age (*FirmAge*), measured as the number of years elapsed between the establishment of the acquiring firm and the patent acquisition

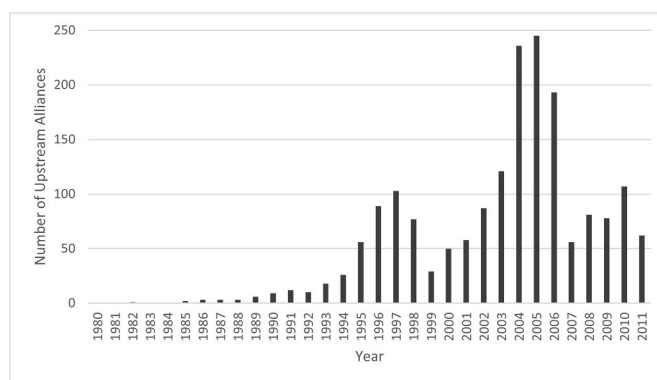


Fig. 2. Number of upstream alliances identified per year.

⁶ The USPTO assigns the following patent classes to the biotechnology domain: 424 (drug, bio-affecting and body treating compositions), 435 (chemistry: molecular biology and microbiology), 436 (chemistry: analytical and immunological testing), 514 (drug, bioaffecting and body treating compositions [different sub-classes]), 530 (chemistry: natural resins or derivatives; peptides or proteins; lignins or reaction products thereof), 536 (organic compounds), 800 (multicellular living organisms and unmodified parts thereof and related processes), 930 (peptide or protein sequence), and PLT (plants).

agreement, since these may affect, respectively, the firm's availability of resources and its experience in the industry. These variables were calculated by relying on data obtained from BioScan, company websites, and web archives such as Orbis and Compustat. We accounted for the presence of non-star scientists (*NonStarScientists*), measured by the logarithm of the number of individuals (plus one) affiliated with the acquiring firm who published at least one scientific article in the patent acquisition year and who do not meet the criteria used for those considered star scientists (Hess and Rothaermel, 2011). In fact, the presence of individuals endowed with scientific knowledge, even though they are not considered stars, may impact the integration and use of externally acquired knowledge. Additionally, we also considered the number of downstream strategic alliances (*Downstream*) established in the five years up to the patent acquisition agreement (Lavie and Rosenkopf, 2006), measured by the number of collaborations involving joint marketing and service, production or supply agreements, since these provide firms with complementary capabilities, know-how, and market knowledge that may, in turn, impact the exploitation of the patents acquired. Moreover, we included a variable controlling for the acquiring firm's technological capital (*TechCapital*), evaluated as the logarithm of the number of patents (plus one) applied for in the five years preceding the patent assignment agreement; this measure represents the knowledge stock that a firm may leverage to innovate (Nooteboom et al., 2007). We also controlled for the type of organization selling the patent (*Assignor*) with a dummy variable taking the value of 1 if the assignor is a firm and 0 (zero) if it is a university or a public research center. Indeed, we believe the content of the technology acquired may have considerably different characteristics depending on the type of organization that developed it (Thursby et al., 2009). Furthermore, we controlled for the firm's experience in acquiring patents (*PrevAcqXP*), measured as the number of patents acquired by the focal firm in the five years preceding the given patent assignment agreement (Leone and Reichstein, 2012). Indeed, the experience of firms in dealing with patent acquisition may influence the effectiveness of this strategy to support their innovativeness. In addition, we used a control variable to further reduce the possibility that patents were acquired for other strategic reasons. Specifically, we verified the technological relatedness of the focal patent to the acquiring firm's patent portfolio, in terms of presence of related patents characterized by the same three-digit U.S. technological classes as the acquired ones (Messeni Petruzzelli et al., 2015a). Then we created a dummy control variable (*TechRelated*), assuming a value of 1 if related patents were included in the firm's portfolio and 0 otherwise. In this way we control for the fact that the focal patents, when related to an acquiring firm's knowledge base, are acquired in order to enhance the innovativeness of the acquiring firm (Wu et al., 2014; Gambardella et al., 2015). Finally, we included year dummies (*Year*) in the model to capture temporal trends (*Year* = 2012 represents the omitted category). Appendix A reports the full description of the calculation and source of data for each variable used in the empirical analysis.

3.3. Analysis

The acquired patent served as the unit of analysis. In our study, the dependent variable (*SelfCit*) is a non-negative integer count variable, which is characterized by overdispersion, since the mean is lower than the standard deviation. The negative binomial model allows for differences between mean and standard deviation, contrary to the Poisson model, which instead assumes an equal value. We then applied the negative binomial model to test our hypotheses (Hilbe, 2007). We based the calculation of significance levels on Huber-White robust standard errors in order to account for heteroskedasticity issues. We tested five different models. Specifically, Model 1 is the baseline model, including only the control variables and the moderators, while Models 2–4 are the partial models, including the linear and quadratic specifications of the independent variable and, separately, each of the moderating variables and the related interaction with the independent variable. Finally,

Model 5 is the full model in which all variables are simultaneously considered. After presenting the descriptive statistics, we focus our results discussion only on the full model, since the tests for potential multicollinearity show the maximum variance inflation factor (VIF) is 6, well below the commonly accepted threshold of 10 (Kleinbaum et al., 1998), thus avoiding the risks of bias. Furthermore, the levels of significance of independent and moderating variables remain consistent across models, thus further reducing collinearity concerns.

4. Results

Table 1 reports descriptive statistics and pairwise correlations with significance levels. All the correlations are below the 0.70 threshold (Cohen et al., 2003), except for the relationships between *Upstream* and *Downstream*, and between *NonStarScientists* and *FirmSize*. However, we ran the models both including and excluding *Downstream* and *FirmSize* and found consistency in our results, thus avoiding concerns about potential bias. Additional descriptive analyses for our sample are provided in Appendix B.

In Table 2, we present the results of the negative binomial models. Model 1 shows that the number of star scientists ($\beta = 0.697$, $p < 0.001$) has a positive significant effect on the extent to which firms internally exploit the acquired technology. The same positive impact is shown by the number of upstream alliances ($\beta = 0.098$, $p < 0.05$), the number of non-star scientists ($\beta = 0.302$, $p < 0.01$), the number of patent claims ($\beta = 0.024$, $p < 0.001$), the number of scientific references ($\beta = 0.003$, $p < 0.001$), the scope of the patent ($\beta = 0.206$, $p < 0.001$), the technological relatedness of the acquired patent with the acquiring firm's portfolio ($\beta = 0.779$, $p < 0.001$), and having a firm as the assignor ($\beta = 0.784$, $p < 0.001$). In contrast, the technological capital of the firm ($\beta = -0.344$, $p < 0.001$), its age ($\beta = -0.004$, $p < 0.01$) and its size ($\beta = -0.053$, $p < 0.05$) exert a negative significant effect. In Model 2, we included the linear and quadratic terms of *PioneeringNature*; in Model 3, we added the interaction between *StarScientists* and the linear term of *PioneeringNature*, while in Model 4 we incorporated the interaction between *Upstream* and the linear term of *PioneeringNature*. Finally, Model 5 is the full model, which is used to test our hypotheses.

Model 5 reveals that Hypotheses 1 and 3 are supported, while Hypothesis 2 is not confirmed. Specifically, for Hypothesis 1 it was shown that the pioneering nature of the acquired patent has an inverted U-shaped effect on the extent to which the acquiring firm exploits it for developing further proprietary technologies. We found statistical support for this hypothesis. In fact, Model 5 shows that *PioneeringNature* is curvilinearly related (inverted-U) with *SelfCit* (*PioneeringNature*, $\beta = -0.018$, $p < 0.001$; *PioneeringNature*², $\beta = -0.0001$, $p < 0.001$). Moreover, in order to provide further corroborating evidence for the inverted U-shaped relationship, we performed the three-step procedure by Lind and Mehlum (2010), as suggested by recent literature (Haans et al., 2016). First, we found that the coefficient of *PioneeringNature*² is statistically significant and consistent with the hypothesized sign. Second, we verified that the slope of the relationship is significantly steep both at the lower bound (value equal to 0.014, $p < 0.001$) and at the upper bound (value of -0.018 , $p < 0.001$) of the *PioneeringNature* measure range. Finally, we calculated the turning point of the relationship and its 95 percent confidence interval, based on Fieller's standard errors method, and found that the confidence interval is entirely located within the data range of our independent variable. Thus, we can be confident that the interpretation of the inversed U-shaped relationship is correct, hence confirming Hypothesis 1.

In Hypothesis 2, we predicted a positive moderating effect of *StarScientists* on the relationship between the pioneering nature of the acquired patent and subsequent internal technological development, implying a shift in the turning point of the relationship (Haans et al., 2016). Investigating the shift of the turning point of the inverted U-shaped relationship, rather than its steepening or flattening, allows to understand how the moderators influence the positive (or negative)

Table 1

Pairwise correlation matrix and descriptive statistics (n = 6208).

Variable	Mean	Standard Deviation	Min	Max	1	2	3	4	5	6
1. SelfCit	1.19	4.70	0	95						
2. PioneeringNature	−14.12	29.80	−410	0	−.11***					
3. StarScientists (log)	0.12	0.28	0	1.67	.08***	.05***				
4. Upstream	0.78	1.58	0	19	.08***	−.02 [†]	.44***			
5. Claims	17.12	15.60	0	99	.10***	−.25***	−.02 [†]	.02 [†]		
6. Scope	1.92	1.01	1	8	.03**	.00	.00	.04***	.12***	
7. PatentAge	17.68	7.63	4	48	−.03*	.26***	.09***	−.02*	−.22***	−.01
8. ScientificCit	20.33	47.98	0	761	.05***	−.43***	−.05***	.05***	.21***	.03**
9. FirmSize (log)	7.33	2.60	.69	11.92	.03**	.07***	.39***	.23***	−.12***	−.04***
10. FirmAge	44.26	49.02	0	192	−.06***	.07***	.20***	−.07***	−.19***	−.12***
11. NonStarScientists (log)	1.25	0.95	0	3.47	.03**	.10***	.60***	.42***	−.06***	.04**
12. Downstream	1.84	3.64	0	27	.10***	−.08***	.33***	.75***	.04**	.07***
13. TechCapital (log)	1.58	0.82	0	4.01	.01	.08***	.36***	.29***	−.11***	−.01
14. Assignor	0.94	0.24	0	1	.04**	−.07***	.06***	.04**	.02*	−.04***
15. PrevAcqXP	13.51	55.16	0	638	.01	.01	.09***	.09***	.00	−.01
16. TechRelated	0.75	0.43	0	1	.05***	−.05***	.05***	.08***	−.02	−.37***
Variable	7	8	9	10	11	12	13	14	15	
7. PatentAge										
8. ScientificCit	−.29***									
9. FirmSize (log)	.27***	−.23***								
10. FirmAge	.13***	−.19***	.49***							
11. NonStarScientists (log)	.23***	−.18***	.74***	.21***						
12. Downstream	−.09***	.09***	.13***	−.09***	.26***					
13. TechCapital (log)	.23***	−.17***	.66***	.31***	.69***	.15***				
14. Assignor	−.16***	.04**	.07***	.15***	−.06***	.00	.06***			
15. PrevAcqXP	.07***	−.03***	.14***	−.03 [†]	.16***	.03**	.12***	.04**		
16. TechRelated	−.09***	.06***	.08***	.03*	.08***	.04***	.30***	.07***	−.01	

†p < 0.10; *p < 0.05; **p < 0.01; ***p < 0.001.

Table 2

Negative binomial regression models.

Dependent Variable: SelfCit	Model 1	Model 2	Model 3	Model 4	Model 5
PioneeringNature		−0.02*** (0.00)	−0.02*** (0.00)	−0.02*** (0.00)	−0.02*** (0.00)
PioneeringNature ²		−0.001*** (0.00)	−0.001*** (0.00)	−0.001*** (0.00)	−0.001*** (0.00)
PioneeringNature x StarScientists (log)			−0.02* (0.01)		−0.02** (0.01)
PioneeringNature x Upstream				0.002* (0.00)	0.003** (0.00)
StarScientists (log)	0.70*** (0.20)	0.67** (0.20)	0.44 [†] (0.23)	0.69*** (0.19)	0.34 (0.24)
Upstream	0.10* (0.04)	0.10* (0.04)	0.10* (0.04)	0.13** (0.04)	0.14** (0.04)
Claims	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)
Scope	0.21*** (0.05)	0.21*** (0.05)	0.21*** (0.05)	0.21*** (0.05)	0.21*** (0.05)
PatentAge	−0.01 (0.01)	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)
ScientificCit	0.003*** (0.00)	0.001 (0.00)	0.001 (0.00)	0.001 (0.00)	0.001 (0.00)
FirmSize (log)	−0.05* (0.03)	−0.08** (0.03)	−0.09*** (0.03)	−0.08** (0.03)	−0.08** (0.03)
FirmAge	−0.004** (0.00)	−0.003** (0.00)	−0.003** (0.00)	−0.003** (0.00)	−0.003** (0.00)
NonStarScientists (log)	0.30** (0.09)	0.35*** (0.09)	0.36*** (0.09)	0.34*** (0.09)	0.35*** (0.09)
Downstream	−0.01 (0.02)	−0.01 (0.02)	−0.01 (0.02)	−0.01 (0.02)	−0.01 (0.02)
TechCapital (log)	−0.34*** (0.08)	−0.37*** (0.08)	−0.38*** (0.08)	−0.38*** (0.08)	−0.40*** (0.08)
Assignor	0.78*** (0.19)	0.68*** (0.19)	0.69*** (0.19)	0.68*** (0.19)	0.71*** (0.19)
PrevAcqXP	−0.01 (0.01)	−0.001* (0.00)	−0.002* (0.00)	−0.002* (0.00)	−0.002* (0.00)
TechRelated	0.78*** (0.12)	0.75*** (0.12)	0.76*** (0.12)	0.76*** (0.12)	0.77*** (0.12)
Patent Class dummies	Included	Included	Included	Included	Included
Year dummies	Included	Included	Included	Included	Included
Constant term	−2.13*** (0.31)	−2.19*** (0.31)	−2.16*** (0.31)	−2.26*** (0.32)	−2.25*** (0.32)
Likelihood ratio test	−6333.68***	−6301.68***	−6299.80***	−6300.14***	−6296.63***
Improvement in fit over the base model		32.00	33.88	33.54	37.05
Number of observations	6208	6208	6208	6208	6208

Huber-White robust standard errors in parentheses. †p < 0.10; *p < 0.05; **p < 0.01; ***p < 0.001.

mechanisms resulting in the main relationship effect (Haans et al., 2016). Hence, discussing the moderation effect, in terms of turning point shift, it is relevant to better comprehend when negative returns set in and, consequently, how the effectiveness (to spur subsequent development) in leveraging the pioneering nature of acquired technologies decreases. Based on this reasoning, Model 5 shows a statistically significant impact of the interaction term, but in the opposite direction compared to our Hypothesis 2 ($\beta = -0.02$, $p < 0.01$), which, accordingly, is not supported. To crosscheck the results, we performed the test suggested by

Haans et al. (2016) to verify whether a translation of the turning point actually occurs, by calculating the derivative of the turning point equation with respect to the moderating variable. We verified that the shift of the turning point is significantly different from zero for meaningful values of *StarScientists* and the effect is consistent with the findings from the main analysis. Therefore, Hypothesis 2 is not confirmed, but we found statistical evidence of the existence of a negative moderating effect. Finally, consistent with our Hypothesis 3, the establishment of upstream alliances has a significant positive moderating effect on the

impact of the pioneering nature of the acquired patent on our dependent variable ($\beta = 0.003$, $p < 0.01$), resulting in a translation of the turning point of the inverted U-shaped relationship. Also, in this case we applied the test proposed by Haans et al. (2016) to show that the shift in the turning point is significant for relevant values of *Upstream* and in the hypothesized direction, thus additionally supporting Hypothesis 3. Table 2 shows that Hypotheses 1 and 3 are also supported in partial models (Models 2–4), while Hypothesis 2 is consistent in partial models (Model 4) and in the full model. Furthermore, we found a significant improvement of the fit between Model 1 and Model 5. Finally, the robustness analyses reported in Appendix C provide additional reliability to our results.

To provide more granular evidence, we plotted the regression results and calculated the maximum values of the inverted U-curves by using coefficient estimates. It is worth to highlight that the following discussion aims at clarifying the regressions results in Table 2, rather than providing additional results. In particular, Fig. 3 depicts the relationship between the pioneering nature of the acquired patent and the firm's subsequent technological developments. The inverted U-shaped pattern shows that an increasing degree of the pioneering nature of the acquired patent provides advantages to the acquiring firm, although only up to a certain threshold. Indeed, after this threshold (−216), the degree of pioneering nature does not offer additional benefits in terms of opportunities to build related proprietary developments, as the curve declines after this point.

Figs. 4 and 5 illustrate the moderating effects of star scientists and upstream alliances. In both cases, we assumed two levels of the moderating variables — low (one standard deviation below the mean) and high (one standard deviation above the mean) — and estimated their effects on the relationship between the pioneering nature of the acquired patent and the extent to which it stimulates the acquiring firm's internal innovation process. Contrary to Hypothesis 2, Fig. 4 shows that by increasing the extent of the pioneering nature of the acquired patent, namely for higher values of the independent variable, firms employing a greater number of star scientists tend to be less effective in internally exploiting the acquired technologies, since the maximum of the curve shifts to the left — from −220 to −307 — as the number of star scientists increases.

Alongside the results of our hypothesis, Fig. 4 also shows that, at any level of *PioneeringNature*, firms employing more star scientists are generally able to more extensively leverage the acquired patent to promote subsequent technological developments. However, their effectiveness in spurring subsequent development based on the acquired patent decreases earlier compared to firms with less star scientists as the

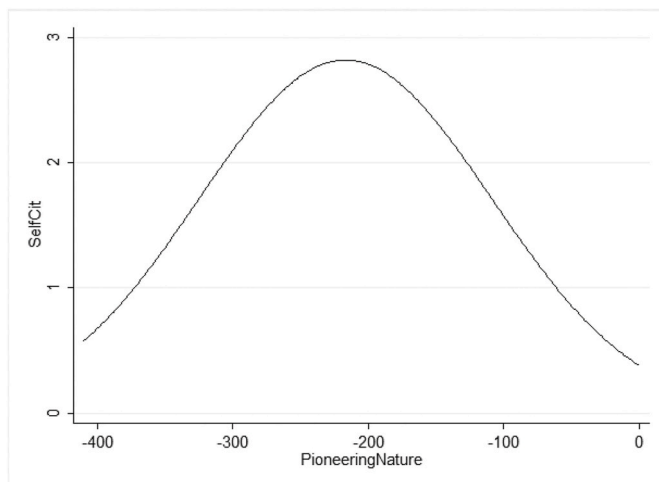


Fig. 3. Pioneering technologies and subsequent internal technological development.

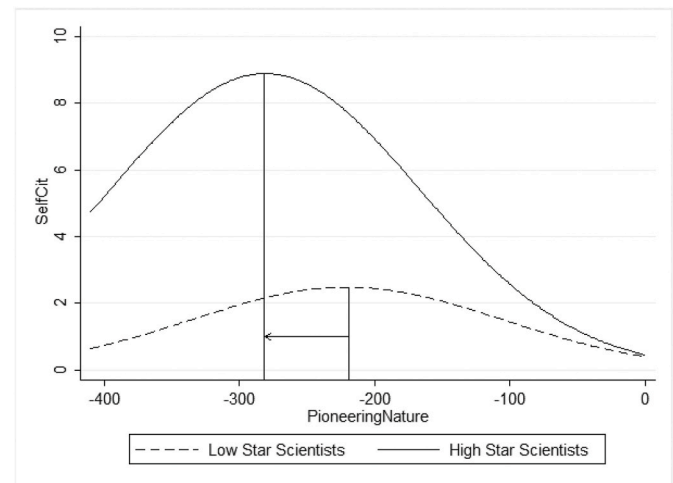


Fig. 4. The moderating effect of star scientists.

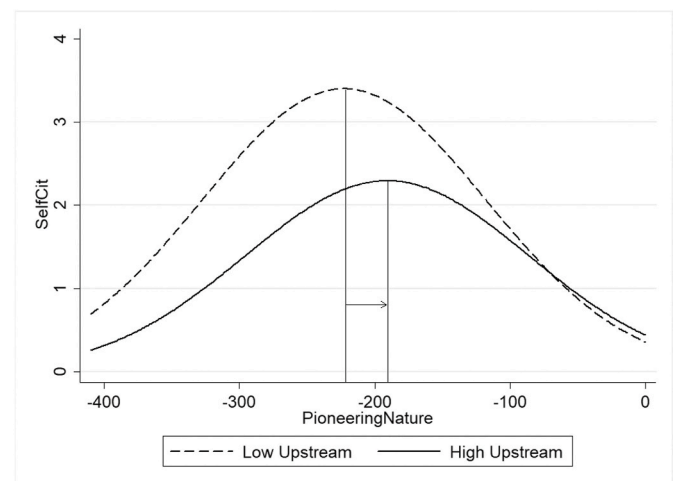


Fig. 5. The moderating effect of upstream alliances.

extent of its pioneering nature grows. The drop of effectiveness is steeper for firms employing more star scientists, to the point that this mechanism becomes barely relevant when firms are called to exploit extremely pioneering technologies. Similar reasoning is applied to Fig. 5, which reveals, consistent with Hypothesis 3, that the threshold marking the point after which the negative returns of the pioneering nature of acquired patents start emerging moves to the right — from −222 to −175 — as firms establish more upstream strategic alliances. What it also emerges from the Figure is that the moderating effect of the upstream alliances results in a higher effectiveness of firms, engaging with more upstream alliances, in leveraging highly pioneering acquired technologies to enhance subsequent development. Accordingly, Fig. 5 shows also that, while, generally, having less upstream alliances allows to more extensively exploit acquired technologies characterized by a smaller extent of pioneering nature, this effect flips for highly pioneering acquired technologies, consistently to what hypothesized.

5. Discussion and conclusions

Pioneering patent acquisition is a relevant and effective boundary spanning mechanism for firms, allowing them to span both organizational and technological boundaries to promote their subsequent technological development (Rosenkopf and Nerkar, 2001). In this paper, we analyzed how firms may exploit the pioneering nature of the acquired

patents to develop technological innovations based on a novel technological trajectory. Specifically, we found that the pioneering nature of the acquired patents has an inverted U-shaped relationship with their subsequent internal exploitation. In fact, the acquisition of technologies with a large extent of pioneering nature provides firms with the opportunity of overcoming competency traps and support their innovativeness (Ahuja and Lampert, 2001). Nevertheless, firms may encounter difficulties exploiting external technologies that have a high degree of pioneering character, due to the lack of appropriate competences (Tushman and Anderson, 1986; Rosenkopf and Nerkar, 2001; Shane, 2001) and absorptive capacity (Cohen and Levinthal, 1990) and the lack of any further contact with the technology originator (the transfer of right is permanently executed in patent assignment contracts). Therefore, as acquired technologies supersede a certain threshold of pioneering nature, their positive impact on the acquiring firms' subsequent internal technological development diminishes.

Furthermore, we argued that, especially in science-based industries, firms may adopt multiple practices affecting their boundary spanning activity (Nickerson and Zenger, 2004; Jacobides and Billinger, 2006; Parmigiani, 2007; Parmigiani and Mitchell, 2009). Indeed, firms may sustain their internal technological development processes by embracing concurrent strategies such as leveraging the capabilities of internal key employees and/or allying with partner organizations (Jacobides and Billinger, 2006; Parmigiani and Mitchell, 2009; Hess and Rothaermel, 2011). Therefore, we focus on the retention of star scientists and experience with upstream alliances since these two factors may support firms in developing "second-order competences" (Dyer and Singh, 1998; Zollo and Singh, 2004) that, in turn, influence the extent to which firms are able to exploit the acquired technologies and create new knowledge through recombination of knowledge across boundaries (Rosenkopf and Nerkar, 2001; Nerkar and Miceli, 2016).

With regard to the retention of star scientists, the previous literature highlighted that employing star scientists has a general positive effect on firms' innovativeness (Zucker et al., 1998; Fleming and Sorenson, 2004; Furukawa and Goto, 2006; Rothaermel and Hess, 2007). However, our analysis showed that the presence of star scientists does not have a direct influence on the exploitation of externally acquired technologies. Actually, due to their status, star scientists may require the autonomy to pursue their own research agenda (Kehoe and Tzabbar, 2015), even undervaluing knowledge coming from others (de Araújo Burchart et al., 2014). This may suggest that star scientists could not be directly involved in the process of recombining the acquired patented solutions, or they have constraints in use them to spur further development.

Additionally, and contrary to our Hypothesis 2, we found that employing star scientists has instead a negative moderating effect on the relationship between the pioneering nature of acquired technologies and the extent to which the acquiring firm exploits them for further developments. Therefore, this finding suggests that while internal star scientists are paramount for building a firm's organizational and technological boundary spanning capabilities (Cohen and Levinthal, 1990; Henderson and Cockburn, 1994; Rothaermel and Hess, 2007), they may become a core rigidity (Leonard-Barton, 1992) when they are called on to exploit the new trajectory defined by the acquired technologies characterized by a remarkable pioneering nature. This is consistent with anecdotal evidence from the target industry. Indeed, biotechnology organizations, traditionally viewed as depositories of the scientific knowledge at the basis of this industry's innovation wave, often give their star scientists freedom and full autonomy in guiding research. This strategy is usually beneficial to organizations, since autonomous star scientists may develop relevant knowledge that can be monetized by organizations even if it is related to non-core fields (e.g. by developing and selling non-core patents). Additionally, through their publications in academic journals, star scientists may increase the status of their organizations, thus providing better access to funds and attracting new talents and collaborations (Gardner, 2017). However, the same circumstances may result in star scientists paying little attention or even

detracting other scientists' attention from leveraging externally acquired technological solutions in place of their ideas. Hence, considering that the effort required to effectively assimilate and recombine external knowledge increases with its pioneering nature, due to its intrinsic novelty with respect to the focal firm's knowledge base (e.g. Lane and Lubatkin, 1998), the retention of star scientists may actually be detrimental for the effective use of pioneering patent acquisition to spur subsequent firm's technological development. This is particularly true, in case when star scientists have attracted economic resources for their own projects, therefore diverting the colleagues' attention from research projects in which star scientists are not (or marginally) involved (e.g. Merton, 1968), such as in the case of an external patent acquisition. This phenomenon could be particularly harmful when the acquired patents are characterized by a notable pioneering nature, since, being novel for the firm, their recombination process requires more resources to be effective (Dahlin and Behrens, 2005; Phene et al., 2006; Dahlander and Gann, 2010). Accordingly, this may reduce the firm's effectiveness in leveraging acquired technologies with a large extent of pioneering nature.

Finally, with reference to experience in upstream alliances, we found it has a positive direct effect on the exploitation of externally acquired technologies. This result may be explained considering that upstream alliances are defined within an overall organizational strategy. Therefore, through alliances, the focal firm can access new and relevant knowledge to be recombined and boundary spanning capabilities that could enrich and be complementary to those already owned, thus constituting a proper base to favor the exploitation of externally acquired patented solutions (Caloghirou et al., 2004).

Furthermore, the experience in upstream alliances positively moderates the impact of the acquisition of highly pioneering technologies on a firm's subsequent developments. Both highly acquiring pioneering technologies and engaging in upstream alliances are boundary spanning activities allowing firms to carry out an explorative search to enrich their knowledge base (Koza and Lewin, 1998). Hence, we point out that by establishing upstream alliances, a firm may encounter the opportunity to develop technological boundary spanning capabilities that make it able to understand, recognize, evaluate, and synthesize technologies owned by partners (Caloghirou et al., 2004). Additionally, as firms gain more familiarity with dealing with upstream alliances, they may be able to reduce the cognitive, transaction, and organizational costs of managing multiple knowledge sources, as well as the related coordination problems and, consequently, further expand their organizational boundary spanning capabilities. This reasoning implies that firms with experience in sustaining upstream strategic alliances may develop "second-order competence", that in turn may increase the effectiveness in re-combination of internal and external technologies showing a greater extent of pioneering nature (Rosenkopf and Nerkar, 2001). Indeed, when the acquired technology is characterized by a remarkable pioneering nature, greater boundary spanning capabilities are required to manage it effectively (Rosenkopf and Nerkar, 2001). Therefore, firms with higher experience in upstream alliances may have developed those capabilities that make them more effective than others in leveraging externally acquired technologies to spur subsequent development, as the extent of their pioneering nature increases.

Concerning the control variables, the analysis shows that some of them influence the organization's exploitation of externally acquired patents. In particular, we point out that the number of claims positively affect the dependent variable. Indeed, this may be explained considering that the higher the number of features of the patented technology the broader its potential application for the firm (Lanjouw and Schankerman, 2004). Moreover, also the scope of the patent positively impacts the number of acquired patent's forward self-citations. In fact, a higher scope of the focal patent provides more technological opportunities to the acquiring firm, that consequently may leverage it to spur subsequent development (Nerkar and Shane, 2007). The number of non-star scientists has also a positive effect on the number of self-citations of the

acquired patents, suggesting that the availability of R&D resources plays a not negligible role in the effective exploitation of externally acquired knowledge (Monteiro et al., 2017). Also acquiring patents from other firms, rather than universities, and the technological relatedness between the acquired patent and the acquiring firm's knowledge base have a positive impact on the dependent variable. Both these results may point out that sourcing knowledge from the same institutional domain and technological similarity may support the acquiring firm in effectively assimilating the acquired technology and exploit it for subsequent development (e.g. Cohen and Levinthal, 1990). Conversely, a number of factors negatively impact the exploitation of externally acquired technology for subsequent development. Concerning firm-related factors, the results suggest that firm size negatively affects the dependent variable. By relating this with the positive impact of having non-star scientists, we may indicate that is not the amount of resources tout court that favors the use of external knowledge, but qualified resources are those making firm effective in exploiting acquired patents (Monteiro et al., 2017). Additionally, firm age and technological capital negatively impact the dependent variable. This could be interpreted considering that older firms and firms with broad technological capital may rely less on externally acquired knowledge to spur further development, since they could already have a core knowledge base developed through years on which develop further innovation (Sørensen and Stuart, 2000). However, they could rely to different knowledge sourcing strategies to spur development. Finally, the previous experience in acquiring external knowledge negatively affects the *SelfCit* variable. We may interpret this result considering that, due to the effort required by this strategy in terms of money and time, companies may prioritize the acquisition of more crucial external knowledge, hence subsequent acquisitions may refer to complementary technologies adding less in terms of marginal value.

5.1. Theoretical implications

This paper provides several theoretical contributions. First, we contributed to the boundary spanning literature, specifically delving into the mechanisms characterizing both technology and organizational boundary spanning (Rosenkopf and Nerkar, 2001; Nerkar and Miceli, 2016). In particular, we studied the acquisition of externally developed technologies characterized by a great extent of pioneering nature as a strategy that allows firms to span their boundaries, by showing how firms develop new technological advancements based on the piece of knowledge acquired. Hence, we provided a better understanding about the advantages and risks of this boundary spanning strategy, that has been partially overlooked in previous literature, despite its increasing practical relevance (Serrano, 2010; Somaya, 2012; Caviggioli and Ughetto, 2013; De Marco et al., 2017; Figueroa and Serrano 2019). In fact, by investigating the acquisition of external technologies through patent acquisition, we analyze a mode that has been scantily discussed in the literature compared to other practices like alliances, corporate venture investments, mergers and acquisitions, and licensing (e.g. Laursen et al., 2010; Van de Vrande et al., 2011). We believe and demonstrate that agreements of this type, involving a permanent change of ownership, actually support the recipient firms in spanning their organizational boundaries, paving the way to subsequent technological evolution and new technological trajectories, which can substantially influence their future competitiveness (Rosenkopf and Nerkar, 2001; Blind et al., 2006; Tsai and Wang, 2008; Mudambi and Tallman, 2010).

Furthermore, by specifically focusing on the pioneering nature of the acquired technology (Ahuja and Lampert, 2001), we discuss the impact of a relevant technology boundary spanning dimension on the subsequent technological development process of the acquiring firms. In so doing, we complement previous studies that investigated the antecedents of the development of pioneering technologies (Van de Vrande et al., 2011) by revealing how the pioneering nature of the acquired technology promotes the spanning of a firm's technological boundaries

and, in turn, its technological evolution. Moreover, while previous works investigated how pioneering technologies affect evolution at industry level (Rosenkopf and Nerkar, 2001), our study digs deeper into the analysis of the causal relationship between the pioneering nature of acquired technologies and technological development itself, by understanding the mechanisms acting at firm level.

In addition, we introduced the analysis of the moderating effect of two complementary search strategies, namely the retention of star scientists and upstream strategic alliances, that contribute to span firm's boundaries. In particular, we show how these two search strategies, while both allowing the focal firm to build boundary spanning capabilities, may have different impacts on the effectiveness of leveraging externally acquired technologies with a great extent of pioneering nature to spur subsequent firm's development. Specifically, our study contributes to and enhances related studies that focused only on one knowledge sourcing strategies (e.g. Wang et al., 2013) by providing a novel grasp on of the interplay between different knowledge acquisition strategies, that may result in distinct effects for the focal firm. With specific reference to the retention of star scientists, considering their role as individuals in the knowledge recombination activity, we also contribute to the emerging of the micro-level theory of knowledge recombination (Fleming et al., 2007; Reinholt et al., 2011; Paruchuri and Eisenman, 2012; Aalbers et al., 2013; Gruber et al., 2013). In fact, we investigate how their productivity and their role affect the effectiveness of the focal firm in leveraging externally acquired highly pioneering technologies to spur subsequent development. Furthermore, by concurrently investigating the focal firm performance when pursuing multiple knowledge recombination efforts, we also contribute to the extant literature dealing with the study of the characteristics, dynamics, and effect of pursuing knowledge recombination to favor technological innovation (Helfat, 1994; Rosenkopf and Nerkar, 2001; Miller et al., 2007).

Finally, we analyzed the relationship between the pioneering nature of the acquired technology and its exploitation by the acquiring firm using three different perspectives of analysis, which are specifically firms, individuals, and alliances, corresponding in our analysis to three distinct strategies aiming at favoring innovativeness. Accordingly, by discussing innovation across multiple levels (Gupta et al., 2007), we grasp notable interaction mechanisms that expand the extant innovation management literature.

5.2. Practical implications

This study also outlines three main managerial contributions. First, while our findings suggest that acquiring technologies showing a great extent of pioneering nature has a positive effect on focal firm's subsequent development, they also warn firm managers and executives of possible side effects of this strategy. In fact, relying upon external technological solutions characterized by a remarkable pioneering nature may not generate the expected benefits. Indeed, acquiring highly pioneering technologies may entail difficulties in assimilating and integrating them into internal development processes. As a consequence, this may limit a firm's effectiveness in exploiting these solutions to enhance their future technological evolution. Accordingly, we suggest managers that acquiring externally developed technologies may notably enhance the technological development of their firms. However, we also point out that as the pioneering nature of the acquired patent increases, their use in firm's innovative processes may be more difficult, with the result that benefits of this strategy may diminish. In turn, our results also show that acquiring externally developed technologies may provide poor benefits even when their pioneering nature is particularly low. In other words, firms can obtain few benefits in exploiting highly incremental external solutions.

Secondly, our results show that employing star scientists has a significant role in moderating the impact of highly pioneering technologies on further advancements. This occurrence may reduce the firms'

opportunities to explore fruitful venues of research by relying upon external technologies, especially when they deviate from the mainstream. Therefore, given the key role of star scientists in a firm’s innovativeness (e.g. [Kehoe and Tzabbar, 2015](#)), we encourage managers to adopt specific human resource practices, such as workshops, seminars, training activities, and incentive systems, to make employees more aware of the potential of external technical solutions that have been acquired and to improve their commitment to leverage the opportunities revealed. Furthermore, managers can also decide to collect advices or even increase the involvement of star scientists in the definition of the company’s innovative strategy, to receive suggestions and to make them more committed towards the diversified strategies pursued. In turn, this may favor star scientists’ commitment when the company acquires externally developed knowledge.

Finally, we noticed the complementarity between acquiring technologies characterized by a remarkable pioneering nature and establishing upstream strategic alliances. Thus, we highlight the idea that upstream alliances may be particularly useful in science-intensive and multidisciplinary industries, such as biotechnology ([Phene et al., 2006](#)). Since they allow the development of organizational and technological boundary spanning capabilities that may support a better exploitation of externally acquired highly pioneering knowledge. Moreover, we point out the advantages that firms can obtain by defining a systemic innovation strategy, to exploit the synergies and mutual reinforcing effects among different knowledge sourcing activities. In fact, broadening a firm’s knowledge base with knowledge from multiple sources may notably increase the effectiveness of the whole innovation process, by exploiting potential complementarities that may emerge.

5.3. Limitations and further research

Of course, this study presents several limitations that may however turn into potential routes for further research. First, we focused on the extent to which acquired technologies are exploited by firms to introduce technological developments, according to their pioneering nature. Developing this insight further, future studies may analyze how firms can capture the value of highly pioneering technologies ([Teece, 1986](#)), for example, by marketing them as new products or processes (e.g. [Nevens, 1990](#)). In this way, future studies may also relate the technology acquisition strategy with the economic performance of the focal firm. Second, further research may also investigate more thoroughly the extent to which the acquisition of patents for defensive reasons or for settling litigation can affect our findings ([Galasso et al., 2013](#)), thus corroborating our analyses and robustness checks. Indeed, in some sectors patent management may aim at different strategic objectives ([Gilardoni, 2007](#)). Hence, digging deeper into the reasons for the patent acquisition, also considering different sectors, may further increase the generalizability of our results. Third, future works could compare the costs and benefits of different types of patent sourcing strategies, namely patent acquisition and patent in-licensing, by evaluating under which conditions firms should prefer one over the other. Actually, the choice between acquisition and in-licensing may affect the extent to which a firm decides to use a patent as the basis for subsequent development ([Figueroa and Serrano, 2019](#)). Therefore, better understanding differences and similarities across the two patents sourcing strategies may

provide relevant insight to increase the effectiveness of the innovation process. Fourth, regarding moderators, it would be interesting to investigate how, in addition to star scientists, retaining star inventors may affect the development of different capabilities and, in turn, affect the influence of the pioneering nature of the acquired knowledge on a firm’s subsequent technological development. Indeed, star scientists may provide mainly basic scientific knowledge that favors the development of boundary spanning capabilities ([Gruber et al., 2013](#)), positively affecting the firm’s innovativeness especially in sectors characterized by high complexity ([Fleming and Sorenson, 2004](#)). On the contrary, star inventors may provide technical knowledge contributing to sustain the innovative productivity of a firm, hence favoring the achievement of a sustainable competitive position ([Kehoe and Tzabbar, 2015](#)). Therefore, they may support the organization in developing patenting capabilities, that in turn affect the investigated relationship in a different way with respect to star scientists. Furthermore, future studies may investigate the effects of star scientists and star inventors outbound mobility and verify how this phenomenon could affect the focal firm’s technological and organizational boundary spanning capabilities. Also, we considered the effect exerted by upstream strategic alliances without investigating the nature of the alliance partner or its characteristics, such as technological and geographic proximity, which may in turn influence processes and outcomes in the alliance (e.g. [Saxton, 1997](#)). Consequently, these characteristics, as well as the specific type of upstream alliance, may be included in future works in relation with the effectiveness in developing different boundary spanning capabilities. Furthermore, the influence exerted by industry-based characteristics, such as market competitiveness, structure, and turbulence, should be accounted for, thereby enlarging the spectrum of investigation and enhancing the multi-level nature of our analysis ([Hitt et al., 2007](#)). Moreover, since patent citations represents flows of knowledge related to the development of the focal patented technology (e.g. [Trajtenberg, 1990](#)), some of them may be more critical than others for the innovation development. However, available patent data do not allow to distinguish the relevancy of patent citations. Further studies may dig deeper into this issue and create a hierarchy of patent citations according to their relevance for the development of the focal technology. Finally, our sample is based on biotechnology firms, which, overall, are an important field in which to analyze the effects of patent acquisition ([Arora and Gambardella, 2010](#)), yet they may present specific industry dynamics and features. Therefore, to further generalize our findings, scholars may extend the analysis to other industrial sectors.

Declaration of competing interest

None.

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Appendix A. Summary of the variables used in empirical analysis

Table A1
Description of the variables used in empirical analysis

Variable	Description	Source of data
SelfCit	Number of forward self-citations of the focal patent from patents filed within five years of the patent acquisition agreement	USPTO
PioneeringNature	Zero minus the number of the focal patent’s backward citations	USPTO
StarScientists (log)		Scopus

(continued on next page)

Table A1 (continued)

Variable	Description	Source of data
Upstream	Logarithm of the number of individuals (plus one) affiliated with the acquiring firm whose three-year moving average (from $t-2$ to t) of scientific articles published is equal to or higher than five in the patent acquisition agreement year (t)	BioScan
Claims	Number of upstream strategic alliances (e.g. research and co-development alliances) established in the last five years up to the patent acquisition agreement	USPTO
Scope	Number of claims reported in the focal patent document	USPTO
PatentAge	Number of distinct three-digit U.S. technological patent classes to which the patent belongs	USPTO
ScientificCit	Number of years between the patent issue year and 2017	USPTO
Patent Class dummies	Number of backward citations to non-patent literature reported in the focal patent document	USPTO
FirmSize (log)	Group of 10 dummy variables assuming the value of 1 if the patent belongs to a related technological class pertaining to the biotechnology field: 424 (drug, bio-affecting and body treating compositions), 435 (chemistry: molecular biology and microbiology), 436 (chemistry: analytical and immunological testing), 514 (drug, bioaffecting and body treating compositions [different sub-classes]), 530 (chemistry: natural resins or derivatives; peptides or proteins; lignins or reaction products thereof), 536 (organic compounds), 800 (multicellular living organisms and unmodified parts thereof and related processes), 930 (peptide or protein sequence), and PLT (plants), or to other technological classes (<i>OtherClasses</i>), or of zero if otherwise	USPTO
FirmAge	Logarithm of the number of firm employees in the year of the patent acquisition agreement	Orbis, Compustat, corporate websites
NonStarScientists (log)	Number of years between the establishment of the acquiring firm and the definition of the patent assignment agreement	Orbis, Compustat, corporate websites
Downstream	Logarithm of the number of individuals (plus one) affiliated with the acquiring firm who published at least 1 scientific article in the patent acquisition year (t) and do not comply with the criteria used to be considered star scientists	Scopus
TechCapital (log)	Number of downstream strategic alliances (e.g. distribution, commercialization, and co-marketing alliances) established in the last five years up to the patent acquisition agreement	BioScan
Assignor	Logarithm of the number of patents (plus one) filed by the firm in the five years up to the patent acquisition agreement	USPTO
PrevAcqXP	Dummy variable taking the value of 1 if the focal patent was acquired from a firm and the value of 0 if the focal patent was acquired from a university or a research center	USPTO
TechRelated	Number of patents acquired by the firm in the five years prior to the patent acquisition agreement	USPTO
Year dummies	Dummy variable assuming the value of 1 if, prior to the acquisition agreement, the focal firm owned at least a patent in the same three-digit U.S. technological patent classes as the acquired one, or of zero otherwise	USPTO
	Group of 31 dummy variables assuming the value of 1 if the patent was acquired in the related year, or of zero if otherwise	USPTO

Appendix B. Descriptive analyses

In order to provide a clearer picture of our database, we performed some additional descriptive analyses. In particular, the descriptive analyses allow to identify temporal trends in patent acquisition and use, also in comparison with internally developed patents. Thus, they provide additional insights that highlight the role of acquired patented technologies for the focal firms in our sample. Moreover, we also analyze if different patent acquisition behaviors are in place according to the size of the focal firm, hence offering a clearer picture of the sample. Figure B.1 shows the temporal distribution of patent acquisitions in our database, thereby highlighting the fact that the investigated strategy has been increasingly adopted by firms since 2001.

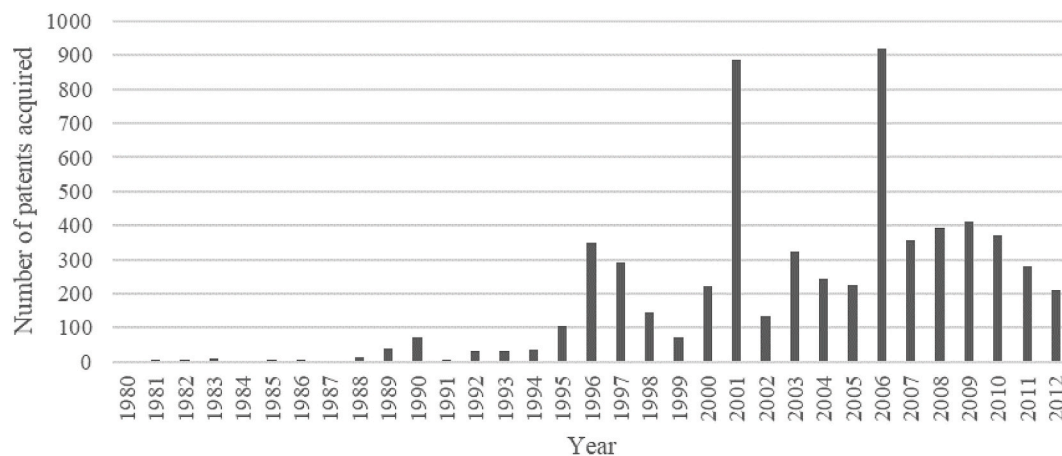


Fig. B1. The temporal distribution of patent acquisition

Additionally, we analyzed the temporal distribution of self-citations (Figure B.2), namely the number of years elapsed between the acquisition of the focal patent and the filing of the acquiring firm's citing patent.

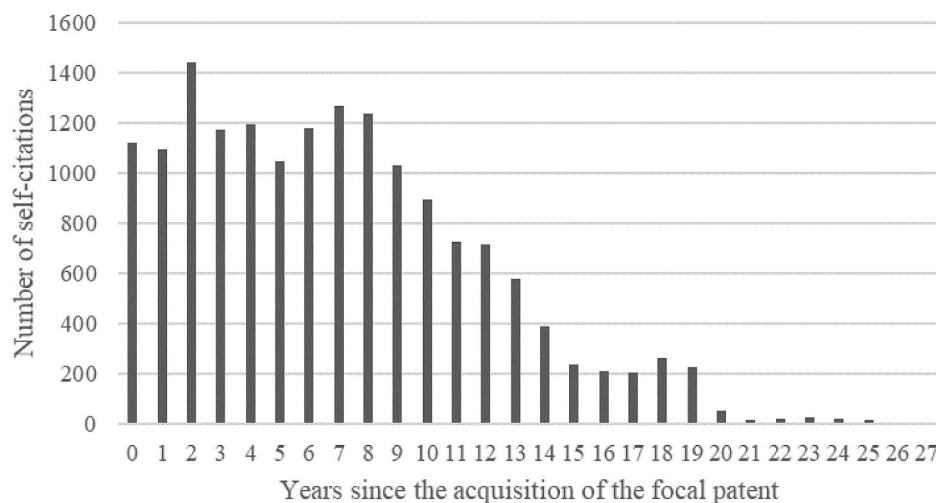


Fig. B2. The temporal distribution of self-citations

Furthermore, we compared the temporal distribution of self-citations between acquired and internally developed patents, in order to spot differences in their use. However, to perform this analysis we needed to use comparable samples to avoid biases. Therefore, we first created a control sample of 4759 patents that were internally developed by the firms in our sample, which includes patents having the same assignee as the focal patent, filing year corresponding to the assignment agreement of the focal patent, and the same main patent class. Then, we removed from the main sample all the patents that did not have correspondence in the control sample, reducing the main sample to 2948 patents. For these two samples we obtained the temporal distributions of the self-citations shown in [Figure B.3](#) and [B.4](#).

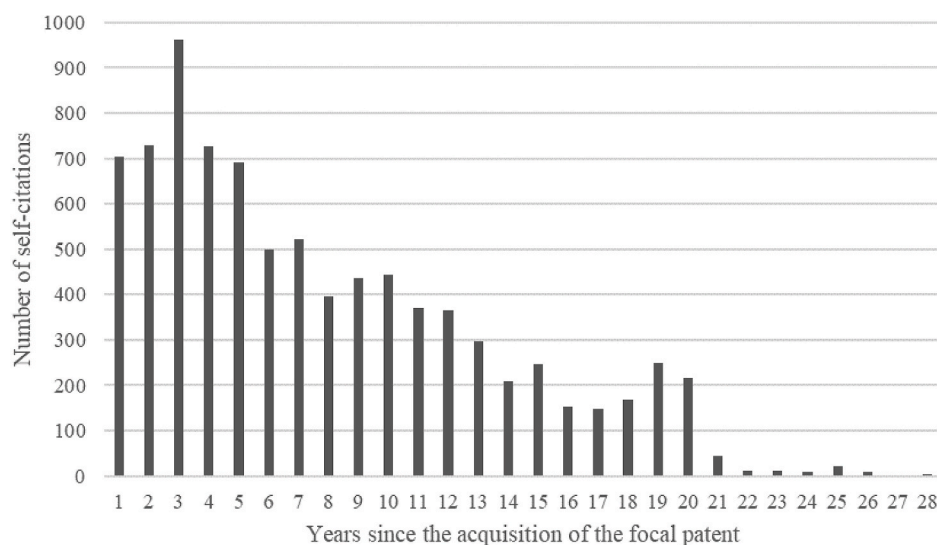


Fig. B3. The temporal distribution of self-citations over time for the main sample

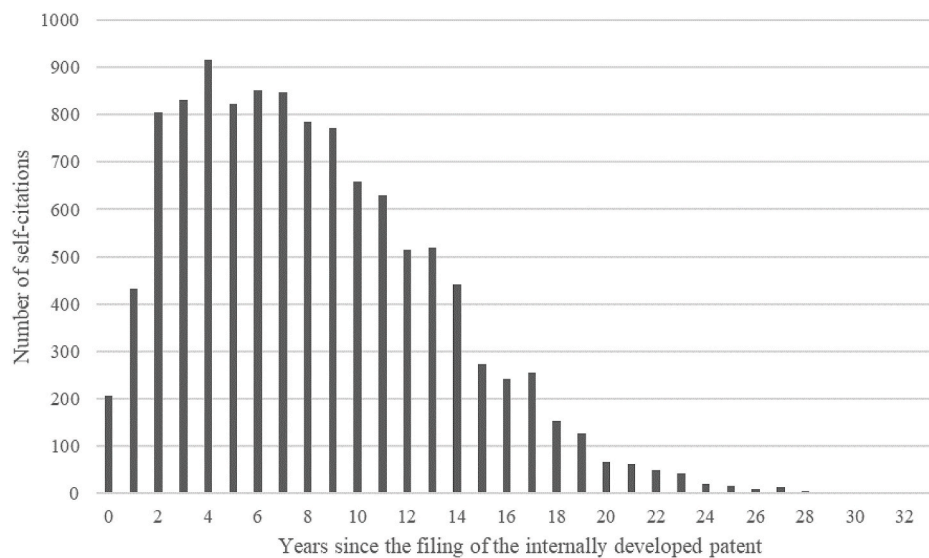


Fig. B4. The temporal distribution of self-citations over time for the control sample

Figure B.3 and B.4 show that acquired patents tend to be exploited earlier compared to internally developed ones, thus suggesting that firms acquire external patents in order to give a boost to their internal innovation processes. Actually, this is consistent with extant literature showing that acquired patents are generally of higher value than internally developed patents, due to an ex-ante selection performed by selling organizations on the basis of patent quality (Gambardella et al., 2007). Furthermore, to confirm this trend, we performed a *t*-test to compare the average number of years it took to self-cite in both cases. The *t*-test provided statistical evidence ($p < 0.001$) of the fact that firms leverage acquired patents earlier than internal patents. Moreover, we also investigated whether there are differences regarding the dependent variable *SelfCit* between acquired and internally developed patents (i.e. if acquired patents are self-cited more or less than internally developed patents). Indeed, an acquired patent is, on average, cited 1.49 times by the acquiring firm's other patents within five years of their acquisition, while an internal patent is self-cited 0.84 times within five years of its filing. A *t*-test revealed that this difference is statistically significant ($p < 0.001$).

Finally, the previous literature highlighted that firms in different size classes may adopt different behaviors and strategies in terms of technology acquisition (Jones et al., 2001). Indeed, large firms may be endowed with greater resources and be characterized by a higher R&D effort, which may increase the effectiveness in leveraging external technologies (Tsai and Wang, 2008). Accordingly, we split our sample into patents acquired by small and medium enterprises (SMEs) and patents acquired by large enterprises in order to perform additional analyses. First, we noticed that large enterprises are the acquirers of 4443 patents in our database, thus representing about the 72% of the sample and showing a higher propensity to acquire patents. Second, we looked for differences in the pioneering nature of the patents acquired. Indeed, large firms tend to acquire more pioneering patents with respect to SMEs. For the former size class, the *PioneeringNature* average of the acquired patents is -12.58 , whereas for the latter size class the average is -17.99 . A *t*-test revealed that this difference is statistically significant ($p < 0.001$). Third, we verified the presence of differences in the use of the acquired patents, by investigating the *SelfCit* variable for the two classes. Our analysis showed that, on average, SMEs cite the acquired patents 1.30 times within five years from the acquisition, while large firms cite the acquired patents 1.14 times in the same timeframe. However, a *t*-test demonstrated that this difference is not statistically significant.

Appendix C. Robustness tests

We performed some additional analyses using different operationalizations of the variables, applying alternative regression models, checking for sampling issues, and executing confirmation tests, in order to check the robustness of our findings. Generally, the robustness tests support the findings in the main analysis, therefore increasing the reliability of our results. In Table C.1 we provide a description with the reference period for the calculation and data source for each variable used in the robustness tests.

Table C1

Description of the variables used in the robustness analysis

Variable	Description	Source of data
Operationalization of the variables		
SelfCit7	Number of forward self-citations of the focal patent for patents filed within seven years of the patent acquisition agreement	USPTO
ShanePio	Number of different three-digit technological classes to which the patents cited by the focal belong, excluding the technological classes to which the focal patent belongs	USPTO
PioneerDist	Number of focal patent technological classes that are not assigned to the biotechnology domain	USPTO
PioneeringSqr	Square root of the number of backward citations of the focal patent	USPTO
StarSci5	Logarithm of the number of individuals (plus one) affiliated with the acquiring firm whose five-year moving average (from $t-4$ to t) of scientific articles published is equal to or higher than five in the patent acquisition agreement year (t)	Scopus
StarSciRH	Logarithm of the number of individuals (plus one) affiliated with the acquiring firm that both published a number of journal articles higher than three standard deviations above the mean of the scientists in our sample and whose publications had been cited more than three standard deviations above the mean of the scientists in our sample	Scopus

(continued on next page)

Table C1 (continued)

Variable	Description	Source of data
Hybrid	Sum of the number of upstream and hybrid (i.e. alliances having both an upstream and a downstream content) strategic alliances established in the last five years up to the patent acquisition agreement	Bioscan
Horizontal	Sum of the number of alliances having (even not exclusively) research and development content and established in the last five years up to the patent acquisition agreement with other biotech firms	Bioscan
VertUpstream	Sum of the number of upstream alliances established in the last five years up to the patent acquisition agreement with non-biotech firms.	Bioscan
CitationPreAcq	Number of citations received by the focal patent from the acquiring firm's patents before the acquisition agreement	USPTO
TechCapAll	Logarithm of the number of patents (plus one) filed by the firm up to the patent acquisition agreement year	USPTO
PioneeringShare	Ratio between the number of internally developed patents with zero backward citations filed in the acquisition agreement year (<i>t</i>) and the total number of internally developed patents filed in the same year	USPTO
PioneeringAvg	Zero minus the average number of backward citations for the internally developed patents filed in the same year as the patent acquisition agreement (<i>t</i>)	USPTO
HiringInventors	Dummy variable assuming value 1 if at least one of the inventors of the acquiring firm's patents citing the focal patent is listed as inventor of the acquired patent itself, and 0 otherwise.	USPTO
Sampling Issues		
Acquired	Dummy variable assuming the value of 1 if the focal patent was acquired, and of zero if it was internally developed	USPTO
ForwardCitations	Number of all the forward citations received by the focal patent up to 2017	USPTO
SubjectAreas	Number of different SIC codes indicating the firm's business activity	SEC filings, Orbis, Compustat, corporate websites
AlliedPatentCitations	Ratio between the number of patents assigned to the firm's alliance partners that were cited by the focal patent and the overall number of the focal patent's backward citations	USPTO

Operationalization of variables

Dependent Variable. We used an alternative operationalization for the dependent variable. Specifically, we used a seven-year fixed timeframe to count forward patent citations (*SelfCit7*) and removed the patents acquired in 2011 and 2012 from the analysis, since the observation interval does not make it possible to calculate the variable for these patents. Even though the operationalization of the variable reduced our sample, we found full consistency with the results in the main analysis.

Independent Variable. We ran the regression model with an operationalization of the pioneering nature of the acquired technologies, as inspired by [Shane \(2001\)](#). Accordingly, pioneering nature is calculated as the number of different three-digit U.S. patent classes in which the patents cited by the given patent are classified, excluding classes to which the focal patent belongs (*ShanePio*). Adopting this measurement, Hypothesis 1 is fully confirmed, while Hypothesis 2 is partially consistent with the main analysis, since the main relationship is negatively moderated by the retention of star scientists, but the coefficient has a lower statistical significance. Moreover, the moderation of the main relationship by *Upstream* is consistent with the main analysis, but with a lower statistical significance. Furthermore, we ran a negative binomial regression not including extremely pioneering patents, namely those patents that do not cite any other patent. In fact, acquiring firms may consider investing in extremely pioneering patents to be akin to taking out an option on a technology with a highly uncertain future that may be exercised or not depending on upcoming opportunities that may emerge ([Hill and Rothaermel, 2003](#)). Therefore, this type of patents may be acquired even though the firm's goal is not to exploit them in the short term. This test reported full support for the results in the main analysis. Furthermore, in line with [Rosenkopf and Nerkar \(2001\)](#), we performed another robustness test in which the pioneering nature of the acquired technology is not measured in absolute terms, as in the main analysis, but as distance with respect to the biotechnology domain. Therefore, we calculated another variable (*PioneerDist*) as the number of the focal patent's technological classes that are not assigned to the biotechnology domain. This test provides full support to Hypothesis 1 and Hypothesis 3, while Hypothesis 2 is not supported. Finally, in order to check for potential biases due to the skewness of the independent variable, we decided to transform it by calculating the square root of the number of backward citations of the focal patent (*PioneeringSqr*). This test fully confirms the results of the main analysis, thus reducing concerns related to the skewness of the *Pioneering* variable.

Moderating Variables. We used two alternative indicators to account for the employment of star scientists. Specifically, we used a more stringent specification of the star scientist definition. Indeed, we considered star scientists only those individuals whose five-year moving average of annual publications, in a specific year, was greater than five in that year (*StarSci5*). With this alternative measurement, results from the main analysis are fully confirmed. Furthermore, we also operationalized star scientists as those individuals having a number of published journal articles higher than three standard deviations above the mean of the scientists in our sample and whose publications had been cited more than three standard deviations above the mean of the scientists in our sample (*StarSciRH*) ([Rothaermel and Hess, 2007](#)). However, unlike the variable included in the main analysis, this operationalization is time-invariant and thus a less reliable representation of the star status of corporate scientists on a yearly basis. Nonetheless, this test also provided full support for the results in the main analysis.

Furthermore, we considered an alternative measure for identifying the upstream alliances. In particular, we performed a test that also included hybrid agreements in the category of upstream alliances, namely those collaborations having both upstream and downstream content established in the five years up to the acquisition agreement (*Hybrid*). With this operationalization, results from the main analysis are fully confirmed.

Moreover, we performed additional tests to confirm that the moderating effect of upstream alliances is related to the development of technological and organizational boundary spanning capabilities and not to other effect related to the partner firm's industry of origin. Therefore, drawing on BioScan database, we highlighted those alliances that focal firms established with other biotech firms ([Rothaermel and Deeds, 2006](#)). Afterwards, we measured a control variable (*Horizontal*) as the sum of the number of alliances characterized (even not exclusively) by research and development content and established in the last five years up to the patent acquisition agreement with other biotech firms. Hence, we ran a regression model assuming *Horizontal* as moderating variable. In this case results of the main analysis are fully confirmed. Moreover, we ran a different regression assuming as moderating variable the number of upstream alliances established in the last five years up to the patent acquisition agreement with non-biotech firms (*VertUpstream*). Also in this case the results of the main analysis are fully confirmed. Accordingly, we can affirm that establishing alliances with a research and development content contributes to the development of technological and organizational boundary spanning capabilities by the focal firm, nonetheless the industry of the partner. This test provides further confirmation to Hypothesis 3.

Control Variables. To account for the possibility that the purchasing firm had already exploited the recombinant potential of the focal technology prior to the acquisition, we added a further control variable to the model, measured as the number of citations received by the focal patent from the acquiring firm's patents before the acquisition agreement (*CitationPreAcq*). In this test, the results from the main analysis are fully confirmed.

Additionally, we measured *TechCap* as the logarithm of the number of patents for which the firm applied up to the acquisition agreement, thus not including the five-year fixed window of the main analysis (*TechCapAll*). With this alternative measurement of the control variable, the main analysis is fully confirmed.

Furthermore, inspired by Van de Vrande et al. (2011), we controlled for the propensity of firms to break away from a local search and develop pioneering technologies. Consequently, we calculated a control variable (*PioneeringShare*) measured as the ratio between the number of patents with zero backward citations that the focal firm applied for in the year of the acquisition agreement and the total number of patents the firm applied for in the same year. With this control variable, results from the main analysis are fully confirmed. Additionally, in order to account for the extent of the pioneering nature of the internally developed patents, we also adopted a different control variable (*PioneeringAvg*), measured as zero minus the average number of backward citations of the patents the focal firm applied for in the same year as the acquisition agreement. In this case as well, the main analysis is fully confirmed.

Finally, in some cases, firms may hire the inventors that developed the focal patent they previously acquired. Actually, extant literature shows that prior inventions of newly hired inventors are more likely to be cited (e.g., Singh and Agrawal, 2011), hence this phenomenon could bias our results. Accordingly, we added a control binary variable (*HiringInventors*) assuming value 1 if at least one of the inventors of the acquiring firm's patents citing the focal patent is listed as inventor of the acquired patent itself, and 0 otherwise. To ensure consistency with the main analysis, we referred to the citations received by the focal patent in the five years following the acquisition. By including this variable in the regression, we found that the results of the main analysis are fully confirmed. Additionally, we point out that the mentioned phenomenon affects about 8% of the patents in our sample.

Estimation models and econometric issues

We ran an alternative negative binomial regression model using clustering across firms with robust standard errors, so as to make the analysis robust vis-à-vis potential correlations among observations regarding patents acquired by the same firm (Hilbe, 2007). We performed this additional test even though our database has unbalanced clusters that may skew the analysis by increasing the reported standard errors (MacKinnon and Webb, 2017). However, in this test as well, the findings from the main analysis are fully confirmed.

Additionally, we also ran a zero-inflated negative binomial model to account for the excess of zero values in the *SelfCit* variable. The resulting findings, again, show full correspondence with the main analysis.

Sampling Issues

In order to check for sample selection bias, we applied the method suggested by Heckman (1979). Consequently, we first created a control sample of 4759 patents that were internally developed by the biotechnology firms, as described in Appendix B. Then we selected, from the main sample, 2948 patents having a correspondence in the control sample. Afterwards, we ran a Probit regression model on the final sample, composed of the main and control samples (i.e. 7707 patents), using the binary variable *Acquired* as the dependent variable. Specifically, the variable *Acquired* assumes the value of 1 if the patent was acquired, while it is 0 if it was internally developed. In this step, we also added two instrumental variables to the model that impact the firm's decision to acquire or internally develop the technology. The first instrumental variable was *ForwardCitations*, which is the number of the forward citations (both self and non-self) received by the patent up to 2017 (e.g. Capaldo et al., 2017), forward citations being a proxy used to estimate patent value (e.g. Trajtenberg, 1990). In fact, patents having higher value may be more appreciated and consequently more likely to be acquired by firms (Fischer and Henkel, 2012). The second instrumental variable was *SubjectAreas*, which is the number of different subject areas in which the firm operates (Diestre and Rajagopalan, 2012). Indeed, firms operating in several subject areas may be less prone to acquire externally developed patents. Their wide knowledge base, spanning over different areas, provides these firms with knowledge and know-how in a variety of fields, thus reducing their need to acquire patented solutions to perform boundary spanning activities (Zhou and Li, 2012). Results from the Probit regression show that only the *ForwardCitation* instrumental variable is statistically significant and in the hypothesized directions. In the second step, we ran a regression on the sample of acquired patents, removing the instrumental variables used in the previous step, while including the inverse Mills ratio calculated from the Probit regression model. Applying this method, we found that the results from the main analysis are fully confirmed.

Furthermore, some firms in our sample may simply decide not to acquire highly pioneering technologies due to the substitutive nature of this strategy compared to the employment of star scientists and the establishment of upstream alliances. Since this phenomenon may introduce bias into our analysis, we performed a test to verify the presence of a further sample selection issue. Therefore, we extracted the highly pioneering acquired technologies from our sample, namely those having a *PioneeringNature* value of zero, and for these calculated the average of the values of *StarScientists* and *Upstream* for the acquiring firm. Then, we calculated the average values of *StarScientists* and *Upstream* for the remaining sample, as well. Afterwards we performed a two-tailed *t*-test to check for differences between the calculated values. The *t*-test showed that there is no statistical evidence to support the existence of a difference between the calculated means of *StarScientists* and *Upstream*. Therefore, we can be certain that there is no difference related to the employment of star scientists and the establishment of upstream alliances between firms acquiring highly pioneering technologies and those not acquiring these types of technology. This contributes to avoiding the risks of bias due to the selection of firms in our sample.

Confirmation tests

We performed additional robustness checks to ensure that the observed relationship between *PioneeringNature* and *SelfCit* is indeed quadratic (Haans et al., 2016). Therefore, we separated the data based on the turning point of the relationship and we performed a linear regression with the subsample below the turning point and a linear regression with the subsample above the turning point. In the former case, we found a positive and significant linear relationship between *PioneeringNature* and *SelfCit*, while in the latter case the linear relationship was negative and significant, consistent with the predicted shape of the relationship. Afterwards, to verify that the shape of the relationship is not defined by a few extreme values, we winsorized our independent variable and ran the regression. The inversed U-shaped relationship is confirmed both with winsorizing at the 5th and 95th percentile levels and with winsorizing at the 10th and 90th percentile levels. These additional tests confirm that the interpretation of the inversed U-shaped relationship is correct.

To give further reliability to the negative moderating effect of the retention of star scientists on the main relationship, we performed an additional test. Specifically, for each acquiring firm's patent that cited the focal patent, we identified the star scientists affiliated with the acquiring firm who are listed as inventors in the citing patent. Then we calculated the ratio between the number of star scientists listed as inventors and the overall number of inventors for the citing patent, and next the average by focal patent of the results was calculated. We verified that only in 33 cases, out of 6,208, was the ratio different from zero, thus confirming the reluctance of star scientists to work on the acquired patents. Moreover, we also verified the tendency of star scientists to develop patents that do not cite the acquired ones. In fact, star scientists developing at least a patent that did not cite an acquired patent were about 55% of the sample, with an average of 10.34 patents registered. This check further confirms that star scientists indeed develop new patents, but without exploiting the externally sourced knowledge.

In a further test, we introduced a control variable measured as the ratio between the number of patents assigned to the acquiring firms' alliance partners and cited by the focal patent and the overall number of its backward citations (*AlliedPatentCitations*). Following this approach, we looked for the possible effects of complementarity or substitutability between the knowledge owned by partners and the knowledge codified in the acquired patent. We ran the regression with this control variable and our results were supported, while the control variable is not highly significant. Therefore, the positive moderating effect of upstream alliances on the relationship between *PioneeringNature* and *SelfCit* does not depend on the link between alliance partners' knowledge and the patent acquired, but rather on the boundary spanning nature of the alliance itself, further reinforcing our results.

Finally, even though biotechnology is a field in which firms tend to acquire patents mainly to nurture their innovation processes, we performed our analysis on those acquired patents that were subjected to follow-up patenting. Thus, we ran the regression model considering only those patents that had at least one self-citation within five years of their acquisition. Results from this test showed full support for Hypothesis 1 and Hypothesis 3, while the interaction between *PioneeringNature* and *StarScientists* is consistent with the main analysis but scarcely significant. Overall, the robustness analyses performed confer additional confidence to our findings.

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